

Review Article

Artificial intelligence in veterinary and animal science: applications, challenges, and future prospects

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ABSTRACT

Artificial intelligence (AI) is increasingly transforming animal and veterinary science through improved decision-making, predictive modeling, and automation. This review comprehensively covers applications of key AI technologies, including machine learning, deep learning, computer vision, natural language processing, robotics, and edge AI, in areas such as disease diagnosis, behavioral monitoring, multi-omics data integration, and precision livestock farming. It also highlights current limitations, including data fragmentation, high implementation costs, and ethical concerns. By addressing these challenges through interdisciplinary collaboration, standardized data systems, and the development of explainable and scalable AI tools, the field can advance toward more sustainable, efficient, and welfare-oriented practices. This review underscores the transformative potential of AI in achieving the goals of One Health and global food security while emphasizing the need for continued research, policy support, and equitable access to AI technologies.

1. Introduction

Artificial intelligence or AI refers to a comprehensive range of technologies that enable machines to imitate human intelligence. This field includes various sub-disciplines such as machine learning (ML), advanced machine learning known as deep learning (DL), natural language processing (NLP), and computer vision (CV) (Goodfellow et al., 2016). These capabilities are revolutionizing various sectors by introducing systems that absorb data, identify trends, and provide decision support with an unprecedented level of accuracy. In the realm of AI, ML and DL have attracted considerable interest due to their strong capacity to manage intricate data and their extensive use in various sectors, such as image and video recognition, forecasting, and automation (LeCun et al., 2015). This development is of great importance for animal and veterinary science, as the data in these fields are often large, disjointed, and difficult to interpret using traditional methods.

As AI techniques have evolved, their integration into the fields of life and agricultural science has brought about significant changes. In the field of animal science and veterinary medicine, where complex, often personalized data is critical, the ability of AI to process and interpret large amounts of data will transform the way professionals and researchers work with animals, and significantly affect animal health, welfare, and productivity (Liakos et al., 2018). AI has shown promise in simplifying complex processes, including the detection of health

problems, the prediction of animal productivity, and the monitoring of behavior, and could potentially provide solutions to the ongoing problems in these sectors.

In the field of animal science, AI is widely used in various fields, which are essential to increase productivity, improve animal welfare, and promote sustainability. A significant application of AI can be seen in the field of animal breeding and genetics, where ML algorithms help predict traits, assess genetic diversity, and improve breeding strategies by interpreting data (Morota et al., 2018). Through better genetic selection, AI helps improve beneficial traits, such as increased growth rates, better feed conversion, and improved resistance to disease. This in turn helps producers meet sustainability objectives and promote animal health (Chafai et al., 2023).

AI plays an essential role in the field of advanced livestock farming, known as precision livestock farming (PLF). PLF uses AI-powered technologies to improve feed management, animal health monitoring, and environmental regulation, minimizing waste and increasing efficiency (Tedeschi et al., 2021). For example, advanced predictive models analyze sensor data to monitor environmental factors and individual animal responses, enabling adjustments to feed, temperature, or hydration that simultaneously promote animal well-being and increase production (Jiang et al., 2023). Using CV and sensor technology, AI helps monitor, in addition to its effectiveness, the behavior of animals. This allows for the real-time monitoring of behaviors such as aggression,

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lethargy, and lameness, which may be signs of health or welfare problems (Congdon et al., 2022).

AI is revolutionizing veterinary diagnostics and treatment planning by providing tools for faster and more precise disease detection. A prominent example is diagnostic imaging, where deep learning models analyze X-ray images, ultrasound images, and CT scans to detect fractures, tumors, or inflammatory diseases with high accuracy and help veterinarians make precise and timely diagnoses. Moreover, by tailoring treatment to the individual health profile of the patient, AI improves therapy outcomes and reduces side effects, thereby promoting personalized veterinary medicine (Akinsulie et al., 2024; Burti et al., 2024).

Predictive models are used in veterinary epidemiology to predict disease outbreaks and guide intervention strategies, and AI applications are also showing promise in this area. AI models can assess outbreak risks by examining data on animal movements, weather trends, and previous disease events. This will help veterinary professionals and authorities take preventive measures (Halasa et al., 2020; Guitian et al., 2023). In addition, AI-powered robotic and automated systems are increasingly being used in clinical and surgical environments to perform repetitive tasks and improve procedural accuracy, particularly in underserved or remote areas (Knudsen et al., 2024).

Regardless of its capabilities, implementing AI in animal and veterinary science faces significant obstacles. A major obstacle is the issue of data quality and availability, as the efficiency of AI largely depends on large, world-class data sets that accurately represent a wide range of animal species. Data inconsistency caused by differences in species, breeds, and environmental conditions can affect the performance and applicability of the model, highlighting the importance of reliable and standardized data collection procedures (Appleby and Basran, 2022; Akinsulie et al., 2024). In addition, there are still issues with the interpretability of AI models, particularly when it comes to clinical diagnosis and decision-making. Unlike traditional statistical techniques, many ML and DL models are complex and often function as “black boxes” that generate predictions without providing precise justifications. When using AI to make decisions that impact the health and welfare of animals, this opacity creates risks, as practitioners may be reluctant to rely on predictions that they do not fully understand or justify (Appleby and Basran, 2022; Hassija et al., 2024). Ethical and regulatory challenges are equally important, as the introduction of AI in animal science often raises privacy, animal welfare, and transparency concerns. Consider automated surveillance, which may lead to data protection issues, in particular about agricultural documents or veterinary health information. Moreover, it is critical to prevent AI technologies from negatively impacting animal welfare or encouraging excessive automation that undermines human oversight in animal care (Coghlan and Parker, 2024; Taji et al., 2024). Despite these obstacles, AI offers transformative potential in animal and veterinary science, addressing limitations in traditional diagnostic and management systems. The ability of AI to process vast amounts of multimodal data, detect early disease indicators, and optimize PLF presents an unprecedented opportunity to improve animal health and productivity. AI-driven technologies, such as ML-based disease prediction models and automated monitoring systems, are already demonstrating superior accuracy and efficiency compared to conventional approaches. This makes AI a valuable tool for improving veterinary diagnostics, disease surveillance, genetic selection, and individualized treatment strategies, ultimately contributing to both animal welfare and agricultural sustainability.

To examine how these technologies affect different aspects of research and practice, this manuscript aims to methodically examine and compile the current applications of AI in animal and veterinary science. To provide a comprehensive picture of AI's impact and potential in these areas, its applications in genetic improvement, precision animal husbandry, behavioral analysis, diagnostic imaging, epidemiology, and tailored veterinary care are examined. This manuscript also discusses current issues and proposes possible solutions, with a focus on data management, model interpretability, and ethical issues. While several

studies have discussed AI applications in veterinary and animal science, there remains a gap in comprehensive analyses addressing the major barriers to implementation. This review uniquely synthesizes recent advancements in AI applications while critically evaluating the specific obstacles limiting their adoption in real-world veterinary practice. By highlighting key challenges such as data limitations, model interpretability, ethical concerns, and regulatory constraints, practical solutions, and future research directions are proposed to facilitate the integration of AI in veterinary medicine and animal science. In line with the goals for animal welfare, health, and sustainable agricultural practices, the current review also seeks to identify future research avenues and interdisciplinary partnerships that can further develop the ethical and successful integration of AI into animal and veterinary science.

2. AI techniques in animal and veterinary science

AI has become a disruptive force in many fields, including veterinary and animal science. The application of AI techniques in this area has the potential to improve research, optimize veterinary procedures, improve herd management, and improve animal care (Akinsulie et al., 2024). This overview highlights the main AI methods that have been used in the field of veterinary and animal science. This section therefore examines the most important AI techniques (ML, DL, CV, NLP, robotics, and edge AI) and their potential applications (Fig. 1).

2.1. ML in predictive modeling and decision-making

ML plays a pivotal role in animal and veterinary science, offering data-driven solutions for predictive modeling, resource optimization, and decision-making. By processing vast amounts of structured data, such as health records, breeding information, and environmental metrics, ML provides valuable insights that enhance disease detection, health monitoring, and resource management. Its adaptability and analytical power make ML an indispensable tool for advancing livestock management, promoting animal health, and achieving sustainable and efficient agricultural practices. Core ML techniques, including supervised learning, unsupervised learning, and reinforcement learning (RL) (Fig. 2), each have their unique contribution to predictive modeling and informed decision-making (VanderWaal et al., 2017; Guitian et al., 2023; Akinsulie et al., 2024).

2.1.1. Supervised learning: Predicting known outcomes from labeled data

Supervised learning is a fundamental ML technique widely used for prediction and classification tasks, particularly when the goal is to determine known outcomes based on input data. This method relies on training models with labeled datasets, where each data point includes input features, such as sensor readings or measurements, paired with corresponding target outcomes, such as disease status or production traits. By learning the relationship between inputs and outputs, supervised learning models can, when applied to new unseen data, make accurate predictions and classifications (Appleby and Basran, 2022; Ghosh and Dasgupta, 2022).

In animal and veterinary science, where labeled datasets are often available from health records, breeding programs, and environmental monitoring systems, supervised learning offers significant utility. It enables the creation of predictive models for tasks such as early disease detection, efficient resource allocation, informed breeding decisions, and productivity improvement (VanderWaal et al., 2017; Guitian et al., 2023). The process of supervised learning typically involves four key stages: First, data collection is conducted to compile a labeled dataset. Each data point consists of input features and a corresponding target label representing the known outcome. For example, in health monitoring, wearable sensor data such as body temperature and activity levels might serve as input features, while the target label could denote disease status, such as healthy or diseased. Next, during model training, the labeled data is used to teach the algorithm to identify patterns and

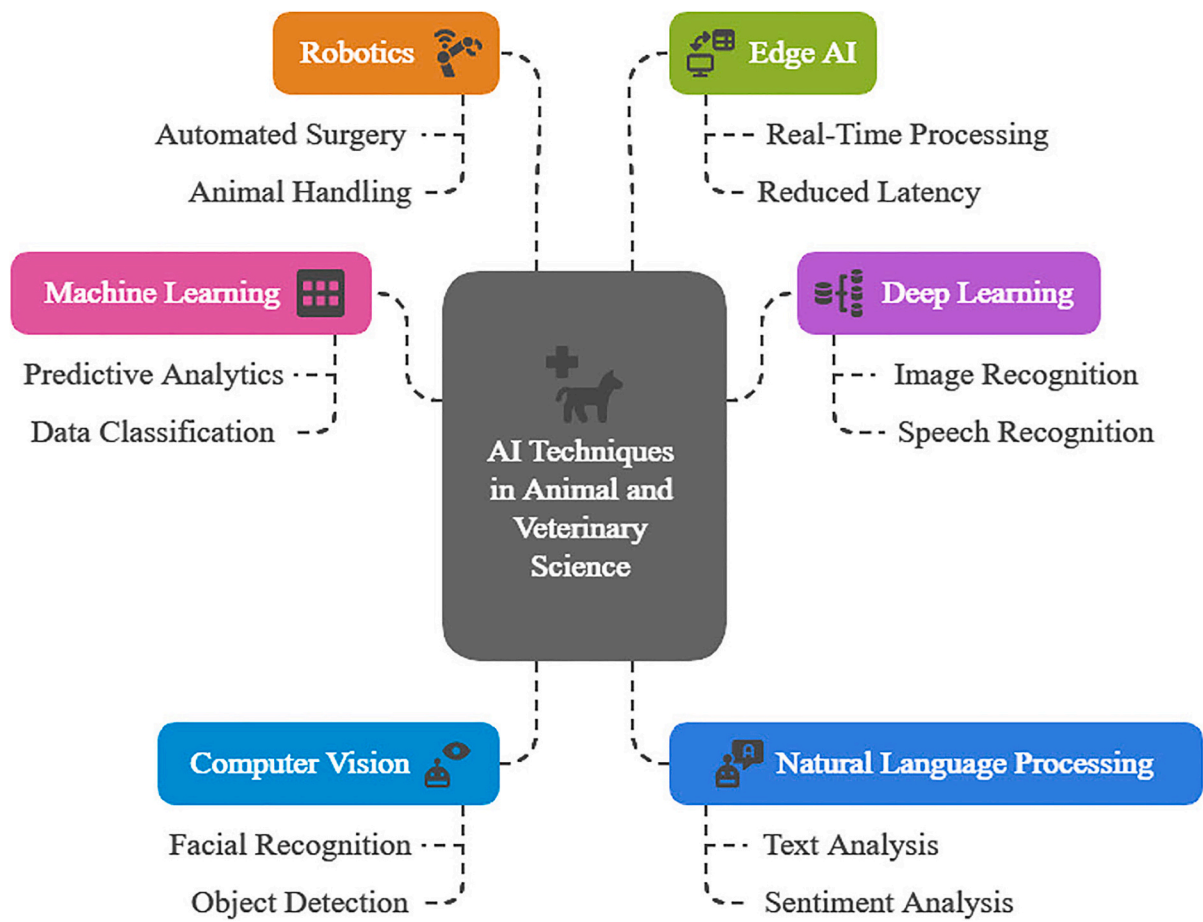


Fig. 1. The most important AI techniques in animal and veterinary science.

relationships between the inputs and outcomes. The model adjusts its internal parameters or weights iteratively to minimize prediction errors, effectively learning to generalize the patterns in the training data. After training, the model undergoes evaluation on a separate testing dataset that was not used during the training phase. This step assesses the model's ability to generalize to new data using metrics like accuracy, precision, recall, and F1-score to measure performance. Finally, the trained model is deployed for prediction, where it processes new data to provide insights. For example, in the field of veterinary diagnostics, it is possible to analyze sensor data and historical health records to predict the likelihood of a disease, allowing for proactive intervention. By offering precise, data-driven insights, supervised learning is transforming animal science and veterinary medicine, driving innovation and efficiency in health management, resource optimization, and productivity enhancement (Russell and Norvig, 2020; Aggarwal, 2023).

Supervised learning has wide applications in animal and veterinary science, particularly in predictive modeling, decision support, and disease management. In animal and veterinary science, several supervised learning algorithms are often used, each with its strengths and suitability for different data types. A description of supervised learning techniques and their applications in animal and veterinary science are presented in Table 1. Also, a list of recent research on the application of supervised techniques in animal and veterinary science is shown in Table 2.

2.1.2. Unsupervised learning: Identifying patterns and groupings without labeled data

Unsupervised learning is an ML approach where models are trained on data without predefined output labels or classifications. The aim is to uncover hidden patterns, structures, or relationships in the data, such as

clusters of similar data points or anomalies. Unlike supervised learning, which uses labeled data to predict outcomes, unsupervised learning focuses on discovering underlying patterns without explicit guidance (Usama et al., 2019).

In animal and veterinary science, unsupervised learning techniques are particularly valuable for revealing hidden relationships in complex, high-dimensional data where labeled examples are scarce or difficult to obtain (Mao et al., 2023). These methods can help researchers, veterinarians, and farmers gain insights into animal behavior, health, genetics, and environmental interactions. This makes them essential tools in areas such as animal welfare, disease monitoring, and breeding programs (Ezanno et al., 2021).

Key concepts of unsupervised learning involve various techniques, with the most common goals being clustering and dimensionality reduction (Usama et al., 2019). Clustering groups similar data points together based on some measure of similarity, while dimensionality reduction reduces the number of variables or features in a dataset while retaining important patterns and relationships. Both goals help uncover the structure of the data, providing insights into previously unknown patterns or groups. Clustering is a popular technique in unsupervised learning, aiming to partition a set of objects (data points) into groups (clusters) based on their similarity (Saxena et al., 2017). In animal science, clustering can be used to identify groups of animals with similar health conditions, behaviors, or genetic traits (Valletta et al., 2017; McVey et al., 2023).

There are several common clustering algorithms used in unsupervised learning. A description of unsupervised learning algorithms in animal and veterinary science is shown in Table 3. Also, a list of recent research on the application of unsupervised techniques in animal and veterinary science is shown in Table 4.

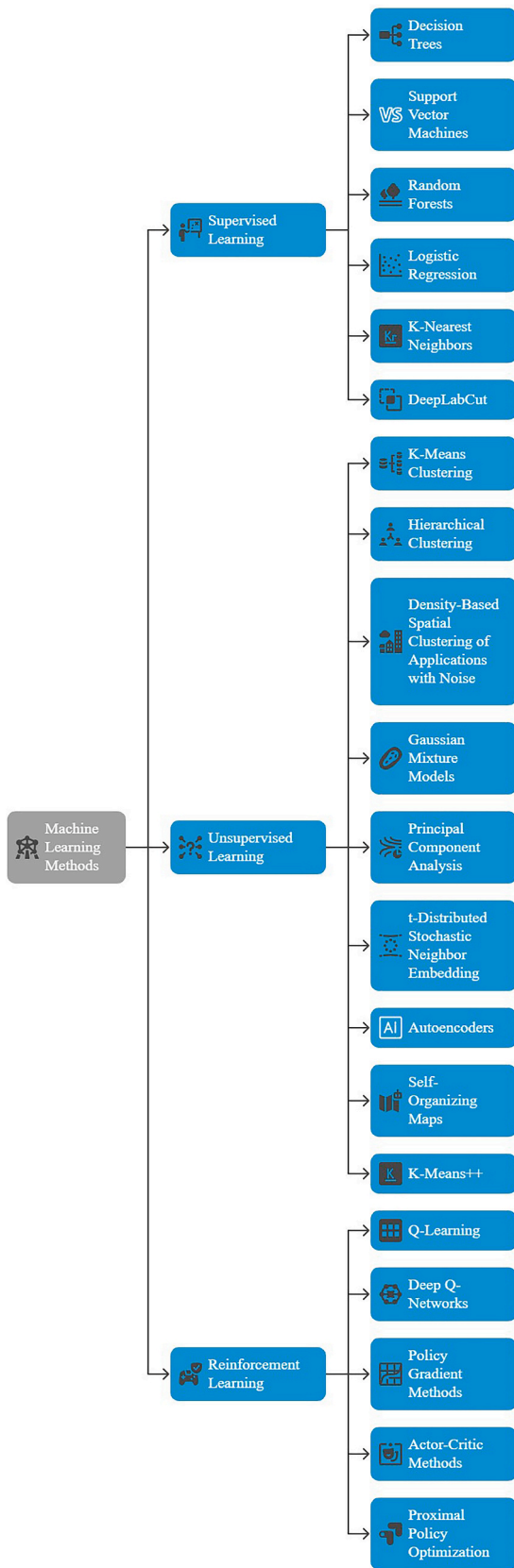


Fig. 2. Major ML techniques in animal and veterinary science.

2.1.3. RL: Optimizing sequential decisions in dynamic environments

The RL is a type of ML in which an agent learns to make decisions by interacting with an environment to maximize a cumulative reward. Unlike supervised learning, which relies on labeled data, RL uses trial-and-error exploration to uncover optimal policies for sequential decision-making. This approach is particularly useful in dynamic environments where conditions change over time and actions have long-term consequences. In RL, the agent observes the environment as it is, acts, and receives feedback in the form of a reward or penalty. The agent gains insights about the best way to proceed by assigning actions to states to maximize the cumulative rewards (Bhavatari et al., 2024). The key components of RL are as follows: Value function (V): An estimate of the expected cumulative reward for a state that leads the agent to long-term benefits; Policy (π): A strategy or rule for selecting actions in certain states; Action (A): Set of possible actions that an agent can perform; Reward (R): Feedback received after an action that indicates the immediate benefit or cost; and State (S): The agent's current situation or context. RL is commonly modeled as a Markov decision process (MDP).

Sequential decision-making is a key feature of RL in dynamic environments. RL works best in situations where decisions must be made sequentially and affect future states and rewards. The agent must also strike a balance between exploration and exploitation, using known actions to maximize rewards and exploring new actions to learn better strategies. In addition, dynamic and stochastic systems that exhibit randomness or change over time, such as animal behavior or environmental fluctuations, are ideal for RL. By capturing these components, RL agents can navigate and optimize complex dynamic environments effectively (Gonzalez, 2024). Numerous RL algorithms have been developed, and each has its advantages: Q-Learning, Deep Q-Networks (DQN), Policy Gradient Methods, Actor-Critic Methods, and Proximal Policy Optimization (PPO). A description of RL techniques in animal and veterinary science is shown in Table 5. Also, a list of recent research on the application of RL techniques in animal and veterinary science is shown in Table 6.

2.2. DL for advanced pattern recognition and decision support

DL is a sub-field of AI that uses artificial neural networks (ANNs) to process and analyze large amounts of complex data, such as how the human brain works. For tasks like image recognition, NLP, and predictive modeling, it eliminates the need for manual feature extraction by allowing machines to learn representations directly from raw data. The basis of DL is, in particular, deep neural networks (DNNs), which consist of several layers of interconnected nodes that transform input data into meaningful patterns (Shafik, 2024). The main advantage of DL is its ability to automatically detect complex structures in large-scale data files, making it an effective tool for many applications, such as bioinformatics, speech recognition, and CV. Various DL model architectures are designed for particular tasks (Fig. 3). Recurrent neural networks (RNNs) are made for sequential data analysis, which makes them appropriate for time-series predictions and language modeling, while convolutional neural networks (CNNs) are frequently used for image-based applications because of their capacity to identify spatial hierarchies in data. Long-range dependencies in sequential data are addressed by variants such as gated recurrent units (GRUs) and long short-term memory (LSTM) networks. In the meantime, autoencoders and generative adversarial networks (GANs) support unsupervised learning tasks like data augmentation, anomaly detection, and image generation. Transformers, a more recent development, have revolutionized NLP and are increasingly being explored in other fields, including veterinary and animal science (Hussain, 2024; Islam et al., 2024). Although DL performs better than conventional ML techniques, it has drawbacks like high processing costs, a requirement for sizable labeled datasets, and an inability to be interpreted in decision-making processes. Efforts are being made to optimize DL models for resource-constrained

Table 1

A description of supervised learning techniques in animal and veterinary science.

ML technique	Potential applications	Description	Advantages	Disadvantages
Decision Trees	- Disease detection - Behavior monitoring - Feeding management	Decision trees use feature values to divide data into branches and, arriving at a leaf node, generate predictions. This framework facilitates sequential, interpretable decision-making.	- Easy to interpret - Handle both numerical and categorical data - Model non-linear relationships	- Prone to overfitting - Sensitive to small data changes - Tends to target features with many levels
Support Vector Machines (SVM)	- Disease classification - Genetic prediction - Stress detection	SVMs use the hyperplane, which best divides classes, to classify data. Using the kernel trick, they can model complicated, nonlinear decision boundaries	- Effective in high-dimensional spaces - Robust to overfitting - Flexible with kernel functions	- Computationally intensive - Requires careful kernel and parameter tuning - Less interpretable
Random Forests	- Resource optimization - Genetic selection - Welfare monitoring	By averaging predictions to reduce variance and increase robustness, random forests combine multiple decision trees trained on arbitrary subsets of data and features	- High accuracy - Less prone to overfitting - Provides feature importance - Handles missing data well	- Computationally expensive - Memory-intensive - Less interpretable than individual trees
Logistic Regression	- Health risk prediction - Disease presence/absence classification	Using a linear combination of input features, logistic regression estimates the probability of an event occurring in binary classification tasks	- Interpretable - Low computational cost - Works well with binary outcomes - Provides probability scores	- Assumes a linear relationship - Sensitive to multicollinearity - Limited for complex relationships
K-Nearest Neighbors (KNN)	- Behavior classification - Disease detection	KNN uses similarities to nearby data points to classify data. For predictions, the most widely used term among nearest neighbors is used	- Simple and intuitive - Effective with small datasets - Non-parametric - Adaptable to different distances	- Sensitive to feature scaling - High memory usage - Poor performance with large datasets
DeepLabCut	- Animal pose estimation - Behavior tracking - Gait analysis	A markerless pose estimation tool called DeepLabCut tracks and labels animal body parts in video footage using deep learning, enabling in-depth behavioral analysis	- High precision in tracking body parts - No need for physical markers - Customizable for various species and poses	- Computationally demanding - Requires large annotated datasets - Sensitive to video quality and lighting

environments and to develop Explainable AI (XAI) to increase their applicability in a wide range of areas (Lamba and Rani, 2024). A description of DL techniques in animal and veterinary science is shown in Table 7. Moreover, a list of recent research on the application of DL techniques in animal and veterinary science is provided in Table 8.

2.3. CV in animal and veterinary science

CV has become a transformational technology in the field of animal and veterinary science, enabling automated, non-invasive, and real-time monitoring of animals using advanced image and video analysis. Disease detection, behavioral evaluation, PLF, and biosecurity improvement are just some of the areas where CV is being used. This technology uses DL algorithms, such as CNNs, to analyze vast amounts of visual data and identify important patterns that can help in decision-making and improve productivity and animal health (Abhiram et al., 2024; Umurungi et al., 2025).

One of the most important applications of CV is in the diagnosis and monitoring of diseases. By facilitating early and accurate disease detection, automated image analysis of medical imaging data, such as X-rays, ultrasound scans, and thermal imaging, has completely changed veterinary diagnostics. AI-driven radiographic interpretation, for instance, helps veterinarians more accurately detect organ abnormalities, tumors, and fractures. Similarly, by examining temperature changes throughout an animal's body, thermal imaging cameras with CV algorithms can identify early indicators of infection, inflammation, or stress. By eliminating the need for time-consuming manual inspections, these non-invasive methods increase productivity and protect animal welfare (Amutha et al., 2024; Ash and Broome, 2024).

Another crucial area where CV is making a big impact is behavioral analysis. CV-powered systems can monitor social interactions within a herd, identify signs of stress, and evaluate an animal's emotional state by

tracking movement patterns, postures, and facial expressions. For example, automated lameness detection in dairy cattle uses CV-based gait analysis, in which AI models examine stride patterns to find anomalies linked to joint or hoof diseases (Ozella et al., 2024). Real-time video monitoring systems are used in poultry breeding to assess flock behavior and identify indications of distress, aggression, or possible disease outbreaks to enable immediate intervention (Campbell et al., 2024).

Moreover, CV is essential to PLF because it makes it possible to automatically identify, track, and monitor each animal individually. Invasive tagging techniques are becoming less necessary as the technology for facial recognition, which was previously used only for human purposes, is being adapted to the identification of farm animals. AI-powered CV systems can identify distinct body shapes, coat patterns, and facial features, enabling personalized breeding management and health monitoring (Egon and Oloyede, 2023). Furthermore, automated weight estimation systems evaluate an animal's body condition using 3D imaging, giving farmers accurate information for growth tracking and nutrition management (Kumari et al., 2023).

CV technology improves fish welfare evaluation and health monitoring in aquaculture. Aquaculturists can take preventative action before the disease spreads by using underwater cameras and AI-driven image analysis to identify abnormalities in swimming behavior, skin lesions, or parasitic infections. Moreover, CV-integrated automated feeding systems optimize feed distribution by assessing fish appetite and activity, which lowers food waste and boosts resource efficiency (Fitzgerald et al., 2025).

In addition, CV is being used to improve livestock production's biosecurity protocols. To stop unwanted access and lower the risk of disease transmission, AI-powered security cameras with object detection features can monitor farm entrances and restricted areas. By identifying contamination, assessing carcass quality, and guaranteeing adherence to

Table 2

A list of recent research on the application of supervised techniques in animal and veterinary science.

Study	ML technique	Objective/Major findings
Benefo et al. (2024)	SVM	This study utilized ML to identify significant <i>Salmonella</i> stress response genes from 177 poultry isolates. Key genes included <i>ybtX</i> , <i>cspA</i> , <i>rpoH</i> , and those involved in lipopolysaccharide biosynthesis, DNA repair, biofilm formation, and cellular metabolism, achieving high model performance.
Chen et al. (2024)	DeepLabCut	This study presented an AI-enabled platform for analyzing animal behavior, enhancing usability through flexible algorithm selection and natural language interaction. A case study demonstrated its effectiveness in evaluating behavioral differences between sick and healthy animals in a medical laboratory setting.
Cihan et al. (2024)	KNN	This is the first study to try diagnosing cardiovascular diseases (CVD) using AI and cattle retinal images. Four DL models and three ML techniques (Extreme Learning Machine, K-Nearest Neighbors, and Support Vector Machine) were employed for their diagnostic potential. This study demonstrated that AI-based system scans significantly improved the accuracy of CVD diagnosis in animals.
Correia et al. (2024)	DeepLabCut	This study compared three methods of DeepLabCut/SimBA, HomeCageScan, and manual scoring for quantifying self-grooming in mice. It found that while DLC/SimBA correlates well with manual scoring for grooming duration, data from both DLC/SimBA, and HCS is less reliable.
Ergin and Koşkan (2024)	KNN	This study compared logistic regression and ML algorithms (KNN, classification and regression trees (CART), and Random Forest) for predicting <i>Fasciola hepatica</i> infection in cattle, revealing accuracies of 0.61 for LR, and 0.57, 0.57, and 0.58 for KNN, CART, and RF, respectively.
Farahnakian et al. (2024)	DeepLabCut	This study investigated the pose estimation of sows and piglets during free farrowing using deep learning. It evaluates five deep learning networks, finding MobileNet most effective, achieving a median test error of 0.61 pixels in detecting seven body parts of the sow.
Farschtschi et al. (2024)	KNN	This study explored the use of digital holographic microscopy (DHM) to identify leukocyte populations in the blood and milk of dairy cows without extensive sample preparation. By isolating different cell types from these samples, the researchers trained ML methods to analyze unknown samples. The findings indicated that DHM is a promising and reliable diagnostic tool for differentiating leukocyte types.
Gomez-Vazquez et al. (2024)	Random Forests	The study utilized PCA to reduce data dimensionality and employed Gradient Boosting and Random Forest algorithms to predict body weight in water buffaloes. Gradient Boosting outperformed Random Forest, indicating its effectiveness in estimating body weight.
Hadar et al. (2025)	Logistic Regression	The study demonstrated that pet insurance data can effectively predict certain cat diseases, such as periodontal disease and skin tumors, using machine learning techniques. The random forest and conditional logistic regression were used for the analysis. The random forest models showed good predictive accuracy, indicating that insurance claims can serve as

Table 2 (continued)

Study	ML technique	Objective/Major findings
Hay et al. (2024)	Random Forests	valuable indicators of health trends in pet populations. The study evaluated four ML models (Random Forest, SVM, CNN, and Multi-Layer Perceptrons) against conventional models for predicting genomic values of growth traits in beef cattle, finding GBLUP is superior in accuracy and model fit.
Imada et al. (2024)	Decision Trees	This study compared decision trees and random forest algorithms to predict future Johne's disease ELISA results using repeat milk tests from 1197 dairy cows, demonstrating the random forest model's superior predictive performance.
Jaddoa et al. (2024)	Logistic Regression	This study developed four ML models: Logistic Regression, Decision Tree, Naive Bayes, and SVM. These models were evaluated based on performance metrics to assess their effectiveness in stress detection and dark meat prediction. The findings contribute significantly to the field of cattle health monitoring by providing a new automated system that enhances the detection of stress and improves meat quality prediction.
Jia et al. (2025)	DeepLabCut	To overcome the difficulties involved in computer vision-based lameness detection, this study suggested a novel cow lameness scoring method based on keypoint localization and temporal information. Keypoint aspects of cow walking were automatically tracked using DeepLabCut, and a model for assessing lameness was developed. The suggested approach successfully tracked important locations in videos with visible light, meeting the need for real-time detection.
Kebede and Megersa (2025)	Decision Trees	This study successfully compared the egg quality traits of Fayoumi and Sasso breeds, revealing significant differences that underscored the importance of breed selection in poultry production. The use of a decision tree algorithm effectively distinguished between the two breeds based on egg weight, shell weight, and yolk weight.
Kittichai et al. (2025)	KNN	This study developed a deep neural network-based image retrieval system for detecting anaplasmosis in cattle, achieving 91.30 % accuracy and 92.83 % specificity. It utilized the Resnext-50 model and KNN optimization to enhance diagnostic efficiency in remote areas.
Küçüktopçu et al. (2024)	KNN	This paper discussed a hybrid approach combining ML algorithms (including KNN) and wavelet transformation for the effective prediction of ammonia levels on poultry farms, which will increase the accuracy of ammonia monitoring and management, which is crucial for animal health and environmental sustainability.
Lund et al. (2024)	DeepLabCut	This study explored using DeepLabCut and Create ML to automate the behavioral coding of African elephants. It compares ML models to manual observations, revealing high accuracy in tracking simple behaviors but limitations in detecting complex behaviors like stereotyped swaying.
Marques et al. (2024)	Random Forests	This study developed predictive models integrating automated activity monitoring and on-farm data, assessing variables like age and environmental conditions. Random forest models excelled in accuracy, enhancing pregnancy prediction success rates compared to using Automated activity monitoring data alone.

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Table 2 (continued)

Study	ML technique	Objective/Major findings
Meghdad et al. (2025)	Random Forests	This study compared six different supervised ML algorithms to identify the most important risk factors for predicting bovine leukemia virus (BLV) seropositivity in dairy cattle in Florida. The Random Forest model emerged as the most effective machine learning algorithm for predicting BLV seropositivity, demonstrating the highest predictive capacity.
Mladenova et al. (2024)	SVM	This study evaluated four ML algorithms, random forest ensemble, decision trees, SVM, and naïve Bayes, using accelerometer and gyroscope data to classify cattle behavior, achieving 88 % accuracy with accelerometer data and 99 % with gyroscope data.
Myint et al. (2024)	SVM	The study developed a real-time cattle lameness detection system using a single side-view camera, employing deep learning algorithms like Mask-RCNN and YOLOv8 for detection, tracking, and lameness classification, achieving high accuracy in monitoring cattle health.
Pirompud et al. (2024)	Logistic Regression	The study applied three ML algorithms (LASSO, classification tree, and random forest) to predict dead-on-arrival (DOA) rates in broiler chickens raised without antibiotics. It identified key variables influencing DOA, such as mortality rates and rearing conditions. This research highlighted the effectiveness of random under sampling (RUS) in improving prediction accuracy, with random forest outperforming other models. This approach aimed to enhance broiler welfare and optimize production strategies by mitigating high DOA percentages.
Poizat et al. (2025)	SVM	To determine the afflicted limb in horses that are trotting in a straight line, this study developed a predictive system using a SVM. The study emphasized how crucial certain factors are for precise classification, like vertical head and withers displacement.
Raza et al. (2024)	Decision Trees	This study revealed that monitoring activity patterns in tropical dairy cows using AfiTag-II biosensors can effectively predict calving days, with random forest and gradient boosting models, highlighting behavioral changes as key indicators.
Risvanli (2024)	Decision Trees	Metrisor is a diagnostic device utilizing gas sensors and ML to detect metritis in cattle. It demonstrated high accuracy, with the random forest algorithm achieving 78.16 % accuracy, and can identify various bacteria associated with metritis effectively.
Shi et al. (2024)	Random Forests	The study utilized wearable inertial sensors to monitor dairy cattle health, employing random forest for behavior recognition and health classification, enhanced by explainable AI to interpret results, demonstrating high accuracy in distinguishing behaviors and identifying health status.
Tun et al. (2024)	Decision Trees	This study presented a top-view depth camera system for cow lameness detection, achieving 99.94 % detection accuracy and 99.92 % tracking accuracy. It emphasizes early detection and utilizes DL for precise classification, significantly enhancing livestock health management.
Van Leerdam et al. (2024)	Logistic Regression	The study developed a predictive model for hypocalcemia in dairy cows using behavioral sensor data and deep learning. It achieved an area under the receiver operating characteristic curve (AUC) of 0.71, identifying cows at risk based on daily

Table 2 (continued)

Study	ML technique	Objective/Major findings
Wang et al. (2024)	SVM	behaviors and additional static features like parity and calving season. The study demonstrated that mixed kernel support vector machine models, particularly SVR_GS, significantly enhance predictive accuracy in genome selection, outperforming traditional models like GBLUP and single kernel functions, while also improving runtime efficiency in processing large sequencing data.
Yan et al. (2024)	Logistic Regression	This study compared 20 ML algorithms for predicting dairy cow respiration rate, identifying CATBOOST with environmental parameters as the best performer. It highlighted the importance of feature selection and identified key factors like air temperature and airflow speed.

hygienic standards, CV protects food safety in meat processing facilities (Petrović et al., 2024).

Despite its many advantages, the implementation of CV in animal and veterinary science poses challenges. Proper lighting, high-resolution cameras, and ideal placement are necessary for high-quality image acquisition, but these may not always be possible in farm settings. The accuracy of AI models can also be impacted by changes in animal posture, fur color, and environmental conditions. Continuous advancements in model generalization capabilities, data augmentation strategies, and algorithm robustness are necessary to overcome these constraints. Future developments will concentrate on expanding real-time CV applications in various agricultural fields, improving edge AI processing capabilities, and integrating multi-modal data sources. By enabling data-driven, automated, and scalable solutions, this technology has the potential to completely transform veterinary care, animal welfare, and livestock management as it develops (Fatoki et al., 2024; Michielon et al., 2024).

2.4. NLP in animal and veterinary science

NLP is an emerging AI-driven technology that allows machines to understand, interpret, and generate human language. NLP is essential to the analysis of large volumes of verbal and textual data in animal and veterinary science. It also helps with decision-making, improves knowledge management, and facilitates communication between researchers, farmers, and veterinarians. Significant progress has been made in automated reporting, literature analysis, voice-assisted systems, and veterinary diagnostics as a result of the integration of NLP with other AI techniques like ML and DL (Geetha et al., 2023).

Information extraction and analysis from veterinary records, research papers, and clinical notes is one of the main uses of NLP in this field. Traditional veterinary documentation contains a large amount of unstructured textual data, which makes manual analysis labor-intensive and inefficient. Algorithms driven by NLP can automatically process these documents, spot important trends, and extract pertinent data regarding genetic markers, treatment results, and disease prevalence. For example, by examining patterns in clinical notes and farmer reports, sentiment analysis of veterinary case reports allows for the early identification of new disease trends. Furthermore, NLP-enhanced search engines help veterinarians and researchers efficiently retrieve relevant scientific literature by understanding the context of queries rather than relying solely on keyword-based searches (Princy Rufina et al., 2023; Davies et al., 2024). In addition, NLP is essential to automated epidemiology and disease surveillance. To identify zoonotic disease outbreaks or livestock infections in real-time, AI-driven text mining tools examine news articles, social media posts, and online forums. NLP models support proactive disease control strategies by detecting references to

Table 3

A description of unsupervised learning techniques in animal and veterinary science.

ML technique	Potential applications	Description	Advantages	Disadvantages
K-Means Clustering	- Grouping animals by health status (e.g., healthy, at-risk, diseased) - Identifying patterns in feeding behaviors	K-Means is a clustering algorithm that partitions data into K clusters based on similarity. It uses Euclidean distance to assign data points to the nearest cluster center	- Simple and efficient - Scalable for large datasets - Easy to interpret	- Sensitive to initial cluster centers - Requires specifying the number of clusters (K) beforehand - Struggles with non-spherical clusters
Hierarchical Clustering	- Classifying animals based on genetic similarity or behavioral patterns - Studying social structure in livestock	Builds a tree-like structure (dendrogram) that shows the hierarchical relationships between data points. It can be agglomerative (bottom-up) or divisive (top-down)	- Does not require predefining the number of clusters - Can handle data with varying cluster shapes	- Computationally expensive for large datasets - Dendrogram interpretation becomes difficult with a large number of clusters
Density-Based Spatial Clustering of Applications with Noise (DBSCAN)	- Identifying unusual behaviors (e.g., stress) in livestock - Detecting anomalous animal movements in wildlife tracking studies	DBSCAN groups data based on density and marks low-density points as outliers. It does not require the number of clusters to be pre-specified and can detect arbitrary-shaped clusters	- Can identify arbitrary shaped clusters - Handles noise well and doesn't require pre-specifying the number of clusters	- Sensitive to distance metric and parameters (eps, minPts) - Struggles with clusters of varying densities
Gaussian Mixture Models (GMM)	- Modeling genetic data to identify subpopulations - Clustering animals based on health status or physiological data	GMM assumes data points are generated from a mixture of Gaussian distributions. It allows for soft clustering, where data points can belong to multiple clusters with different probabilities	- Can model complex relationships in data - Handles soft clustering, where data points belong to multiple clusters	- Computationally expensive - Assumes data follows a Gaussian distribution, which may not always be the case
Principal Component Analysis (PCA)	- Reducing the complexity of behavioral data from sensors or cameras - Analyzing genetic variation in breeding programs	PCA is a linear dimensionality reduction method that transforms data into principal components to explain the most variance in the dataset, reducing the dimensionality	- Reduces complexity by retaining most of the variance - Helps visualize high-dimensional data in 2D or 3D	- Linear method; may not capture non-linear relationships - Interpretation of principal components can be difficult
t-Distributed Stochastic Neighbor Embedding (t-SNE)	- Visualizing genetic diversity in livestock populations - Clustering behavioral traits in animals for welfare studies	t-SNE is a non-linear dimensionality reduction technique that preserves the local structure of data, making it effective for visualizing high-dimensional data in lower dimensions	- Effective for visualizing high-dimensional data in 2D or 3D - Captures complex, non-linear relationships	- Computationally intensive for large datasets - Output visualization can be hard to interpret compared to PCA
Autoencoders	- Extracting features from genomic data to identify important traits - Analyzing movement patterns in animals from video data	Autoencoders are neural networks used for unsupervised learning that reduce dimensionality by compressing data into a smaller encoding and reconstructing it back to the original form	- Can handle non-linear relationships - Flexible and adaptable to a variety of data types (e.g., images, genomic data)	- Requires large amounts of data to train effectively - Computationally expensive and complex to interpret
Self-Organizing Maps (SOM)	- Visualizing behavioral data patterns in livestock - Mapping environmental impacts on animal health	SOM is a type of neural network that projects high-dimensional data onto a lower-dimensional grid. It's useful for visualizing complex relationships in data	- Useful for visualizing high-dimensional data - Captures complex relationships in data and allows for the interpretation of clusters	- Requires careful tuning of network size - Interpretation of the resulting map can be difficult
K-Means++ (Enhanced K-Means)	- Grouping animals by disease risk based on physiological data - Segmenting animal populations for genetic studies	K-Means++ is an improved version of K-Means that better initializes cluster centers to achieve more accurate clustering	- Reduces the initialization problem of K-Means, improving performance - Simple and efficient	- Still sensitive to the number of clusters (K) - May not work well with non-spherical clusters

disease symptoms, geographic locations, and risk factors. Moreover, NLP-powered chatbots and virtual assistants are being used in veterinary telemedicine to improve accessibility to veterinary knowledge, especially in remote areas, by offering immediate advice on common health concerns, medication dosages, and biosecurity protocols (Sruthy, 2024).

NLP is being incorporated into voice-activated farm management systems for livestock farming, which allow farmers to communicate with AI-powered helpers using natural speech. Through the processing of spoken commands and the provision of practical recommendations, these voice-based systems offer real-time insights into animal health, feeding schedules, and environmental conditions. Without having to manually enter data, farmers can, for instance, request feed adjustments, receive weather-related alerts, or ask about the health status of their herds, enabling smooth farm operation management (Jayalakshmi et al., 2024; Kazembe and Mkandawire, 2024).

Furthermore, NLP makes it easier for farmers and veterinarians working in various linguistic regions to receive multilingual translation

services. AI-driven language models can automatically translate research findings, prescription instructions, and veterinary guidelines into a variety of languages, facilitating international knowledge sharing and bridging communication gaps. This ability is especially helpful in One Health initiatives and the global livestock trade, where efficient communication between various regions is crucial (Joshi et al., 2024).

Automated report generation for farm analytics and veterinary diagnostics is another noteworthy use of NLP. By summarizing patient histories, test results, and treatment plans, AI-powered NLP systems relieve veterinarians of administrative duties and facilitate more effective clinical decision-making. To guarantee uniform documentation procedures in veterinary clinics and animal health research, these systems can also produce structured reports from unstructured textual data (Boguslav et al., 2024).

NLP has the potential to revolutionize animal and veterinary science, but it also faces some obstacles. Developing highly accurate language models is hampered by domain-specific jargon, inconsistent clinical

Table 4

A list of recent research on the application of unsupervised techniques in animal and veterinary science.

Study	ML technique	Objective/Major findings
Albechaalany et al. (2024)	Hierarchical Clustering	The study analyzed beef quality from 15 cattle breeds by examining physiological traits, sensory characteristics, and muscular features. It identified distinct breed characteristics, revealing that muscular traits positively influence sensory quality, while physiological traits negatively impact it.
Brock et al. (2021)	SOM	This study presented a novel method for categorizing herd types in livestock systems by fusing machine learning (SOM) with expert knowledge. This strategy is used in Ireland's cattle industry, where ongoing discussions on surveillance and control for endemic cattle diseases can benefit from a thorough understanding of herd types.
Cho et al. (2024)	Autoencoders	This paper proposed a weakly supervised representation learning method for mastitis classification in dairy cows using a 1D-convolutional autoencoder. It achieves an F1-score of 0.6, demonstrating effective performance with low-cost labels from milking data.
de Oliveira Bezerra et al. (2024)	K-Means Clustering	The paper explored cluster analysis in Nelore cattle by correlating visual scores with measurements, using the k-means algorithm to cluster cattle based on weight and visual scores.
Dubourvieux et al. (2023)	DBSCAN	Cumulative Unsupervised Multi-Domain Adaptation (CUMDA) enhanced Holstein cattle Re-ID by accumulating knowledge across multiple unlabeled domains, improving specialization in known environments while maintaining performance in source domains, thus addressing challenges in identity diversity and farm appearance variations.
Gomez-Vazquez et al. (2024)	PCA	Please see Table 2
Inadagbo et al. (2024)	K-Means Clustering	This paper presented an ML approach using CNN and YOLOv8 to quantify dairy cow behaviors, specifically drinking frequency and brush use, enhancing precision in assessing thermotolerance and aiding ranchers in mitigating heat stress effects on milk production.
Lashin et al. (2024)	PCA	To better align treatment strategies with disease progression and optimize intervention timing, this study moved the focus from traditional disease detection or diagnosis to refined severity classification of mastitis in Holstein-Friesian cows using an interdisciplinary approach that integrates advanced machine learning methods within a veterinary context. To preserve important features and improve computational efficiency, dimensionality reduction was applied using PCA and correlation-based feature selection in both Random Forests and Deep Neural Networks.
Lee et al. (2024)	Autoencoders	This paper proposed a generative active learning framework using a variational autoencoder to generate radiology data, addressing data scarcity in computer-aided diagnosis (CAD) systems in veterinary medicine. It demonstrates improved data quality and classification performance through data augmentation and effective training strategies.

Table 4 (continued)

Study	ML technique	Objective/Major findings
Li et al. (2023a)	K-Means++	The YOLOv5-KCB algorithm improved individual pig detection by utilizing optimized K-means clustering with 1-IOU, CA attention mechanisms for improved feature extraction, and BiFPN for effective feature fusion, achieving superior accuracy in recognizing pig heads and faces.
Liang et al. (2024)	DBSCAN	This paper presented an automatic method using YOLOv8m-seg for detecting group-housed pigs, employing a spatial moment algorithm and agglomerative clustering to monitor movement efficiently, achieving over 1500 h of video analysis with high accuracy and low processing time.
Liu et al. (2024)	Hierarchical Clustering	This paper presented a two-step framework utilizing unsupervised machine learning to analyze cow behavior from GPS tracking data, segmenting movement into clusters and classifying them into walking, grazing, and resting behaviors without requiring human observations.
Liu et al. (2025)	DBSCAN	A fattening pig grouping system based on 3D point cloud weight estimation was created in this study by combining cloud servers and edge devices, designing hardware, and developing algorithms to create a weight estimation model and an application platform. Clustering is necessary on the point cloud following statistical filtering to eliminate background and filter noise points. DBSCAN clustering was applied in this study.
Michelena et al. (2024)	PCAT-SNE	Using smart monitoring collars to gather data from dairy cows in intensive and expansive farms, this study concentrated on the analysis of animal behavior. The behavioral patterns of the two types of farms are compared in this paper using dimensional reduction techniques like PCA, Isomap, and t-SNE. The findings show that these methods are capable of successfully differentiating behaviors under various management regimes.
Michelena et al. (2025)	K-Means Clustering	Using daily activity data captured by a commercial monitoring collar, this study assessed and contrasted three unsupervised anomaly detection techniques to determine estrus in dairy cows from intensive farms. This study shows that these methods can be very helpful in farm management, offering insightful data, boosting output, and ultimately boosting the farm's financial performance.
Mor et al. (2024)	Hierarchical Clustering	A dichotomic questionnaire was used for association analyses, hierarchical clustering, and PCA in a cross-sectional study to find infectious cattle that were diagnosed using a direct technique.
Pegoraro et al. (2025)	PCA	By connecting hematological and biochemical analytes with cutaneous lesion patterns, this study used PCA to find predictive indicators of disease progression in cats with sporotrichosis, proving the utility of PCA in veterinary diagnostics and disease prediction.
Reis et al. (2024)	PCA	Using Fourier-transform infrared spectroscopy on blood serum, this study identified high-fertile Nelore female cows for artificial insemination with 100 % accuracy using ML algorithms, which could enhance the results of artificial

(continued on next page)

Table 4 (continued)

Study	ML technique	Objective/Major findings
Sabarish et al. (2024)	Hierarchical Clustering	insemination and reproductive management techniques. PCA successfully divided groups according to molecular data. V3-V4 sections of 16S rRNA gene profiles were constructed using milk samples from a Jersey cow with mastitis and a healthy cow as a control. The diversity and abundance of bacteria were then compared. The 16S rRNA dataset yielded 16,000 sequence numbers in total, with a cumulative length of 250 base pairs. The overall view of the discovered relationships between the bacterial populations in the milk samples was shown by the hierarchically clustered heat map.
Ser et al. (2025)	Hierarchical Clustering	The performance of CNN, ConvLSTM, and VGG19 algorithms was compared in classifying sheep based on behavioral reactivity. For this purpose, the scale test, one of the fear tests that trigger stress-related behaviors in sheep, was applied. Sheep were divided into two clusters “active” and “passive” according to their reactivity, using unsupervised clustering algorithms on both manually obtained data sets and motion index data sets developed with dense optical flow.
Skjerve et al. (2024)	DBSCAN	This paper employed density-based clustering with noise and fuzzy c-means to clean and segment livestock data, analyzing 65,446 records based on age at slaughter, carcass weight, and average daily carcass gain, resulting in three distinct production system clusters.
Smit et al. (2023)	SOM	Triaxial accelerometers were attached to the collar and harness of twelve healthy domestic shorthair cats. Accelerometer and video data were simultaneously gathered over seven days. The summarized accelerometer data was matched with 24 behaviors that were annotated from the video recordings. Random forest and SOM techniques were used to create models for every mounting location. These findings suggest that cat-specific behaviors can be detected by triaxial accelerometers.
Tran and Quach (2023)	Autoencoders	This paper proposed the identification of lesions utilizing Autoencoder and YOLOv6 models for disease classification and detection in chicken flocks. These methods could be used to identify disease carriers in captive poultry populations early and ensure quality feed for humans.
Tselios et al. (2024)	t-SNE	This study explored advanced thermal imaging combined with statistical texture analysis and t-SNE for detecting subclinical mastitis in dairy sheep, achieving 84 % classification accuracy and demonstrating 80 % sensitivity and 92.5 % specificity when correlated with California mastitis test results.
Tsuda et al. (2022)	SOM	SOM was utilized to analyze spatial patterns of wildlife-vehicle crashes, revealing distinct relationships between crash locations and land use classes. Most collisions occurred near forests and water bodies, while fewer incidents were found near sugarcane fields and urban areas.
Turek et al. (2025)	K-Means Clustering	It was proposed in this study that quantitative measurement of the relative bone mineral density (rBMD) of the long

Table 4 (continued)

Study	ML technique	Objective/Major findings
Valente et al. (2024)	t-SNE	pastern bone of horses using conventional radiographic and metal density standards might be useful in the evaluation of bone biology in the field of horses. The rhabdomyolysis of the long pastern bone varies in healthy horses between cortical (CB) and trabecular (TB) thickness. The k-means clustering algorithm successfully distinguished these types of bones from those radiographs taken with different settings of the X-ray tube. This study developed a heart-CNN algorithm to predict the severity of myxomatous mitral valve disease using thoracic radiographs, classifying dogs based on American College of Veterinary Internal Medicine and Mitral Insufficiency Echocardiographic grading systems, achieving high accuracy values indicating good accuracy for different disease stages. Please see Table 2
van Leerdam et al. (2024)	K-Means Clustering	This study used the Deep Embedded Clustering technique to examine the possibilities of unsupervised DL for automatic equine gait classification. This method is contrasted with other supervised and unsupervised algorithms as well as different pre-processing methods. Results showed that probabilistic-based clustering algorithms, such as GMM, perform better in situations where cluster densities vary significantly, which is common in the field of animal activity recognition.
Waele et al. (2024)	GMM	This study presented a multi-view fusion system using three Zed2 cameras to automatically measure cattle body sizes, achieving high accuracy with average relative errors of 2.32 % to 5.22 % compared to manual measurements, addressing challenges in traditional measurement methods.
Wu et al. (2024)	DBSCAN	

documentation, and variability in veterinary terminology. Furthermore, to train AI models, NLP applications need big, high-quality datasets; however, compared to human medical data, access to annotated veterinary texts is still restricted. To overcome these constraints, linguists, AI researchers, and veterinary specialists must work together to improve language models designed especially for the animal health industry (Boguslav et al., 2024; Sun, 2025).

The future advances in NLP will focus on improving contextual understanding, integration of multi-modal data sources (such as combining text with images or sensor data), and increasing the capabilities of conversational AI in real-time. With further development of these technologies, NLP will be a vital tool for veterinary science as it will allow for more effective data analysis, improve the accessibility of veterinary knowledge, and support decision-making processes in animal health management and veterinary care.

2.5. Robotics in animal and veterinary science

In animal and veterinary science, robotics has quickly become a novel technology that offers automated solutions for a variety of tasks, such as precision farming, biosecurity, veterinary surgeries, and animal health monitoring. The need for increased precision, labor reduction, and efficiency in animal handling is what is driving the growing integration of robotics into livestock management systems. Robotic milking

Table 5
A description of RL techniques in animal and veterinary science.

ML technique	Potential applications	Description	Advantages	Disadvantages
Q-Learning	- Studying the effects of heat stress on animals - Energy conservation monitoring networks - Livestock behavior experiments	A value-based RL algorithm that learns the optimal action-value function (Q-value) to determine the best action for maximizing cumulative reward in each state	- Simple to implement and interpret - Effective for discrete action spaces	- Struggles with large or continuous state-action spaces - Requires extensive exploration, which can be computationally expensive
Deep Q-Networks (DQN)	- Automating activity monitoring using video data - Optimizing feeding rations in livestock systems	Extends Q-Learning using deep neural networks to approximate the Q-value function, enabling learning in high-dimensional state spaces such as video or sensor data	- Handles high-dimensional data effectively - Suitable for complex environments with large state-action spaces	- Computationally intensive - Requires a large amount of training data
Policy Gradient Methods	- Adjusting environmental conditions (e.g., temperature, humidity) for optimal animal welfare - Controlling robotic systems in animal handling	A policy-based RL algorithm that learns the optimal policy directly by maximizing expected rewards, suitable for continuous or complex action spaces	- Works well with continuous action spaces - Avoids the need to store a Q-table, making it memory-efficient	- Convergence can be slow - May require careful tuning of hyperparameters for stability
Actor-Critic Methods	- Balancing resource allocation in livestock farming - Improving energy efficiency in livestock systems	Combines value-based and policy-based approaches, using an actor to select actions and a critic to evaluate the action's value, balancing exploration and stability	- Balances efficiency and stability - Effective for large or continuous action spaces	- Computationally expensive due to simultaneous learning of actor and critic - Requires careful design to prevent instability during training
Proximal Policy Optimization (PPO)	- Optimizing handling procedures to reduce stress in animals - Simulating behavioral experiments for animal training	A modern RL algorithm that improves policy gradient methods by ensuring small updates to policies, making them more stable and robust	- Provides stable training - Handles both continuous and discrete action spaces effectively	- Computationally intensive for large-scale applications - Requires significant data and resources for training

systems are among the most well-known uses; they have revolutionized the dairy industry by enabling cows to be milked autonomously and at their own pace. In addition to increasing milk production by lowering animal stress, these systems provide real-time milk composition monitoring and early disease detection for conditions like mastitis. Another crucial element of automated dairy and poultry farming is robotic feeders, which optimize nutritional intake and minimize waste by tailoring feed rations to the needs of individual animals (Bhoj et al., 2024; Kaur, 2024; Lokhorst and Norton, 2024).

Robots are essential for disease detection and health monitoring in addition to feeding and milking. In livestock housing facilities, automated mobile robots with sophisticated sensors, such as gas detection and thermal imaging, continually evaluate the surrounding environment. By spotting possible stressors like elevated ammonia levels, inadequate ventilation, or extremely high temperatures, these robots enable farmers to take prompt corrective action (Sneha et al., 2024; Haribabu et al., 2025). To prevent serious health problems, robots equipped with cameras and AI-driven image analysis software are also used to identify lameness, abnormal gait, or deterioration of body condition in cattle and poultry. This allows for early intervention (Neethirajan et al., 2024). Robots are used in poultry farms to sort chicks, gather eggs, and monitor bird behavior for indications of stress, aggression, or disease (Yang et al., 2024). By eliminating the need for human interaction with animals, such automation greatly improves biosecurity and lowers the risk of disease transmission.

Robotics on the ground and in the air are transforming herd management in large-scale livestock farming. Drones with AI-powered image recognition software can identify missing or injured animals across large rangelands, track animal movement, and check the quality of pastures. To minimize labor costs and guarantee the best possible pasture utilization, ground-based robotic shepherds have been developed to autonomously guide livestock to designated grazing areas (Camacho et al., 2024; Ali et al., 2025). By automating processes like animal handling, carcass sorting, and quality control, robotic systems are also enhancing meat processing and slaughterhouse operations by lowering worker fatigue and raising standards of hygiene (Kim et al., 2023).

Robotic advancements have also had a significant positive impact on veterinary medicine, especially in surgery. More accuracy is made

possible by robotic-assisted surgical systems in intricate operations like minimally invasive laparoscopic surgery, orthopedic interventions, and tumor excisions. These tools improve a surgeon's vision and dexterity, which improves surgical results, shortens recovery periods, and lowers the chance of complications. Moreover, through controlled movement exercises, robotic rehabilitation devices help injured or post-surgical animals regain their mobility (Buote, 2024; Borah et al., 2025).

Robotics in animal and veterinary science has many benefits, but it also has drawbacks, such as high implementation costs, the requirement for specialized technical knowledge, and opposition from conventional farmers who might be reluctant to use automated solutions. Furthermore, a crucial area of research continues to be making sure that robotic systems can adapt to various farm environments and livestock species. Future developments will concentrate on integrating AI-driven decision-making for autonomous farm management, increasing machine-animal interaction to guarantee livestock welfare, and making robotic solutions more affordable and scalable (Essien and Job, 2024; Lokhorst and Norton, 2024). There is no doubt that the continued development of robotics will be essential for PLF and veterinary care in the future, opening the door to more efficient and sustainable animal management systems.

2.6. Edge AI in animal and veterinary science

Edge AI is a quickly developing technology that allows real-time data processing without depending on cloud-based infrastructure by delivering AI capabilities straight to local devices. This development is especially helpful in the fields of veterinary and animal science, where quick decisions are necessary to ensure biosecurity, optimize farm operations, and monitor health. By integrating AI-powered analytics into hardware like smart cameras, sensors, and wearables, edge AI improves response times, lowers latency, and protects data privacy, in contrast to traditional AI models that rely on distant servers for computation (Pasupuleti, 2024; Prangon et al., 2024; Srinivasagan et al., 2025).

Real-time monitoring of health through wearable biosensors is one of the most important applications of edge AI in livestock production. Intelligent tags, ear tags, and implanted sensors equipped with AI algorithms continuously monitor physiological and behavioral parameters

Table 6

A list of recent research on the application of reinforcement learning techniques in animal and veterinary science.

Study	ML technique	Objective/Major findings
Ali et al. (2024a)	PPO	This paper explored using Proximal Policy Optimization (PPO), a deep reinforcement learning algorithm, to enhance battery management in dairy farming, achieving a 1.62 % improvement in reducing electricity import from the grid, thus promoting energy efficiency and sustainability.
Ali et al. (2024b)	Q-Learning	This research employed Q-learning to develop an effective policy for battery management in dairy farming, addressing challenges like electricity demand variability and renewable generation stochasticity. The results show significant reductions in electricity costs compared to traditional baseline algorithms.
Bak et al. (2017)	Policy GradientMethods	Policy gradient methods can optimize animal training protocols by inferring learning rules from behavioral changes and selecting stimuli that maximize learning rates, as demonstrated in the study, which combines reinforcement learning with adaptive optimal experimental design for efficient training outcomes.
Chappidi et al. (2024)	DQN	This study introduced an integrated approach using Graph Neural Networks (GNNs), advanced Q Learning, Multi-Attribute Utility Theory (MAHP) with deep learning, and Generative Adversarial Networks (GANs) for data augmentation. The model employed DQN for adaptive decision-making. This research advanced automated wildlife monitoring, which is crucial for biodiversity conservation and addressing ecological challenges through integrated computational techniques.
Chen et al. (2023)	DQN	This paper proposed a DQN-based method to model fish school behavior, using individual states defined by directional angles and a reward function based on neighbor changes, demonstrating effective collective behavior through training in single and multi-learner scenarios.
Katahira and Kimura (2022)	Actor-Critic Methods	This paper explored the actor-critic learning model's ability to explain choice behavior, highlighting its unique interaction between past rewards and choices, which differs from Q-learning, thus providing insights into optimizing animal behavior through reinforcement learning principles.
Khan et al. (2025)	PPO	This paper presented the application of advanced route optimization techniques in the dairy industry, particularly through the use of Deep Reinforcement Learning (DRL) with PPO. The findings suggested that implementing DRL-PPO can lead to significant cost savings and improved profitability for logistics companies handling perishable goods. This positions the model as a strategic opportunity for transforming logistics practices within the dairy sector.
Malte et al. (2023)	PPO	The combination of Graph Neural Networks (GNNs) and Deep Reinforcement Learning (DRL), specifically the PPO algorithm, has been shown to significantly reduce the damage caused by infectious disease outbreaks in animal trade networks, achieving up to a

Table 6 (continued)

Study	ML technique	Objective/Major findings
Mohandas et al. (2022)	Q-Learning	35 % reduction in the estimated average number of infected animals compared to naïve testing strategies. To determine the context of the dog barking pattern based on multiple real-time scenarios, such as whether the dogs are alone, staring at strangers, or displaying an eagerness to fight, an ML algorithm is used to analyze the barking patterns of two distinct dog species: Rajapalayam Hound and Kombai Hound.
Shuvaev et al. (2020)	Actor-Critic Methods	This paper discussed R-learning within an actor-critic model as a biologically relevant mechanism for sequential decision-making, linking reinforcement learning, the marginal value theorem, and Bayesian inference to inform stay-or-leave decisions in dynamic environments. This study proposed a methodology using deep reinforcement learning to enhance decision-making in marine ranching, focusing on risk and disaster management. It includes a disaster simulator and applies reinforcement learning algorithms to optimize responses to randomly generated disasters.
Song et al. (2023)	Actor-Critic Methods	This study offered a thorough strategy to improve dogs' well-being using a variety of cutting-edge technologies, including a system that creates customized meal plans using reinforcement learning based on the breed, age, weight, health, and emotional state of the dog.
Wimukthi et al. (2023)	Q-Learning	An environmental perception Q-learning (EPQL) approach to extending the lifespan of poultry farm monitoring networks was considered in this study. In general, other models could be used in place of the implemented temperature and humidity model of heat stress for poultry to increase its range of applicability, according to the EPQL framework.
Wu et al. (2021)	Q-Learning	This paper discussed the use of DQN in visual tracking, enhancing robustness and accuracy in challenging scenarios. This approach can be adapted for animal activity monitoring systems, improving tracking reliability and efficiency in processing video data of animal movements.
Yang and Huang (2019)	DQN	This paper proposed an energy-adaptive monitoring system for smart farms using solar sensors on cows, employing uncertainty-aware Deep Reinforcement Learning (DRL) to ensure high-quality data collection despite energy fluctuations and cyberattacks, with Multi-Agent PPO (MAPPO) showing the best performance in monitoring quality.
Zhang et al. (2023)	PPO	The research focused on developing an environmental control model for confined broiler houses using reinforcement learning, specifically the Q-Learning algorithm. It addresses the challenges of poor environmental conditions by implementing a multi-terminal environment acquisition system based on NB-IoT technology for real-time data collection. This approach enhances stability and uniformity compared to traditional threshold switch control methods, effectively regulating various environmental factors within the chicken house, thereby improving overall conditions for meat chickens.
Zhang et al. (2024b)	Q-Learning	

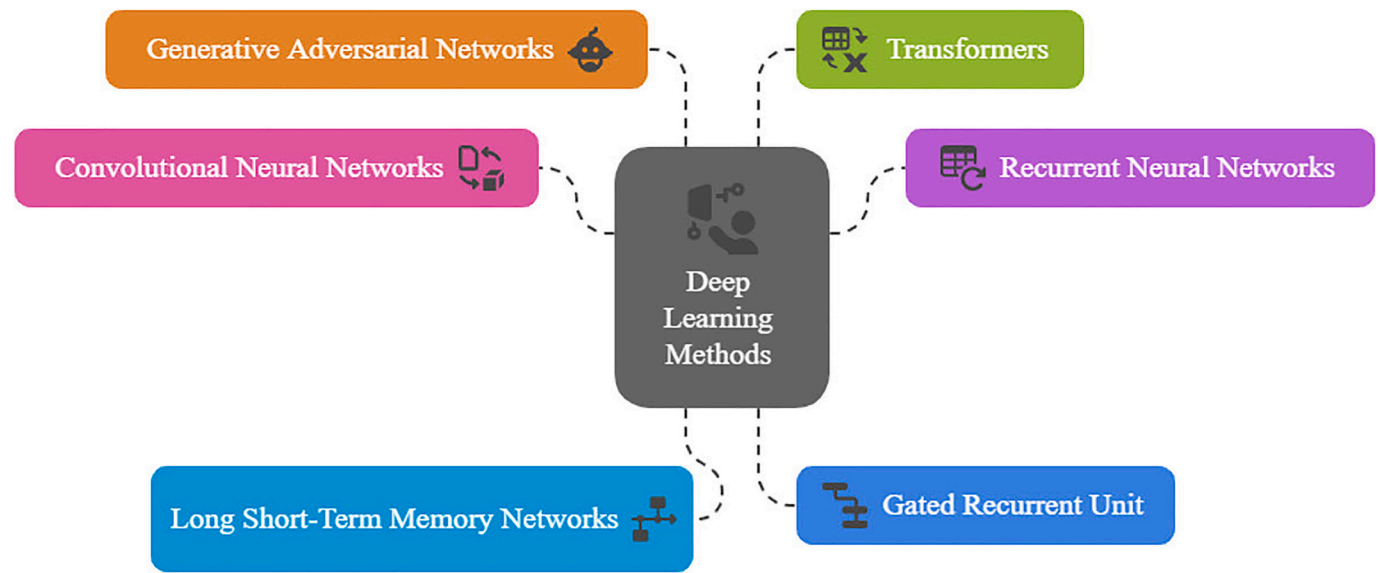


Fig. 3. Major DL techniques in animal and veterinary science.

Table 7
A description of DL techniques in animal and veterinary science.

DL technique	Potential applications	Description	Advantages	Disadvantages
Convolutional Neural Networks (CNNs)	- Medical imaging - Disease detection - Automated classification of livestock conditions	CNNs use convolutional layers to extract spatial features from images, making them ideal for analyzing veterinary radiographs, ultrasound scans, and skin lesion detection	- High accuracy in image recognition - Automated feature extraction	- Requires large labeled datasets - Computationally intensive
Recurrent Neural Networks (RNNs)	- Behavioral monitoring - Disease outbreak prediction - Speech and sound analysis in animals	RNNs process sequential data by retaining memory of past inputs, making them useful for time-series analysis in veterinary and animal science	- Effective for sequential data - Suitable for temporal pattern recognition	- Prone to vanishing gradient problem - Struggles with long-term dependencies
Long Short-Term Memory (LSTM) Networks	- Animal movement tracking - Stress detection - Prediction of reproductive cycles	LSTMs are a specialized type of RNN designed to handle long-range dependencies in sequential data	- Solves vanishing gradient issue - Retains long-term dependencies	- Computationally expensive - Requires extensive hyperparameter tuning
Gated Recurrent Unit (GRU)	- Animal movement tracking - Wearable sensor data analysis - Anomaly detection	A simplified version of LSTM that uses gating mechanisms to control information flow	- Less complex than LSTM - Faster training while maintaining efficiency	- May not capture long-term dependencies as effectively as LSTM
Generative Adversarial Networks (GANs)	- Synthetic data generation - Anomaly detection - Improving image datasets	GANs consist of two networks (generator and discriminator) that compete to produce high-quality synthetic data, useful in data augmentation for veterinary applications	- Effective for generating realistic synthetic data - Enhances model training	- Training instability - Risk of mode collapse
Transformers	- Text analysis in veterinary literature - Disease prediction - Genomic data interpretation	Transformers use attention mechanisms to process entire sequences efficiently, allowing for improved accuracy in analyzing complex datasets	- High performance on sequential data - Better parallelization	- Requires large datasets - Computationally expensive

such as heart rate, breathing rate, body temperature, and movement patterns. They detect deviations from normal patterns immediately and trigger warnings of early signs of disease, heat stress, or changes in reproductive status (Narang et al., 2024; Yu et al., 2024). In dairy farming, for example, edge AI monitoring systems can predict the onset of mastitis by analyzing the milk and its conductivity, enabling immediate intervention before the appearance of clinical signs (Tian et al., 2024). Similarly, in beef production, wearable AI devices detect stress indicators and help farmers optimize harvesting practices to reduce cortisol levels and improve meat quality (Džermeikaitė et al., 2023; Arora et al., 2024).

Through the processing of data from smart sensors placed in barns,

poultry houses, and aquaculture systems, Edge AI also plays a critical role in environmental monitoring. By monitoring temperature, humidity, air quality, and noise levels, these sensors dynamically modify the surroundings to guarantee the best possible care for the animals (Jebari et al., 2023). To reduce respiratory issues in confined animal housing, for example, automated ventilation systems with edge AI integrated can control airflow based on real-time CO₂ and ammonia concentrations (Cho et al., 2025). In addition, to ensure ideal conditions for the production of fish and shrimp, AI-enhanced water quality sensors in aquaculture farms analyze parameters like pH, oxygen levels, and toxin presence (Huang and Khabusi, 2025).

Edge AI drones and autonomous robots are revolutionizing grazing

Table 8

A list of recent research on the application of DL techniques in animal and veterinary science.

Study	DL technique	Objective/Major findings
AlZubi (2024)	CNN	This study underscored the potential of CNNs in early lumpy skin disease detection in cattle, paving the way for practical applications in veterinary medicine.
Atitallah et al. (2025)	CNN	This paper evaluated CNN models for classifying horse health and identifying conditions like tiredness and lameness. It demonstrated advancements in veterinary science through DL, enhancing diagnosis accuracy and efficiency by analyzing horse images and mobility patterns.
Das et al. (2024)	Transformers	This study investigated transformer models for predicting pathogen presence in pastured poultry farms, enhancing prediction accuracy through attention mechanisms and feature importance analysis. It emphasized explainability and the integration of farm management practices with microbiome data for improved food safety.
Durairaj et al. (2024)	LSTM	This paper utilized LSTM to predict cow lameness by analyzing health parameters, demonstrating its practical application in veterinary science for monitoring and diagnosing animal health effectively.
Laleye and Mousse (2024)	RNN	This paper demonstrated the application of RNNs in the automatic recognition of laying hen vocalizations, enhancing behavior monitoring and health assessment in poultry farming, thus advancing research and practical applications in animal and veterinary science.
Mehta and Saini (2024)	CNN	This paper demonstrated the application of CNNs in diagnosing cattle diseases, specifically mastitis, foot and mouth disease, and bovine respiratory disease, achieving a classification accuracy of 95 % through a hybrid model with SVM, showcasing significant advancements in veterinary diagnostics.
Moutaouakil and Falih (2024)	RNN	This paper demonstrated RNNs' effectiveness in classifying cattle behavior through time series data, enhancing automated monitoring in livestock. This advancement supported improved animal welfare, productivity, and management practices, showcasing RNNs' potential in veterinary science and animal behavior research.
Oh et al. (2024)	CNN	This paper highlighted the application of a deep prototype alignment network (DPANet) for few-shot segmentation in veterinary medicine, specifically for accurate heart and left atrial enlargement segmentation in canine chest radiographs, showcasing advancements in CNN-based methodologies in veterinary diagnostics.
Otsuka et al. (2024)	CNN LSTM	This paper highlighted the application of CNNs in wild animal behavior classification, demonstrating their superior performance over classic ML methods. It emphasized advancements in model architecture, including a four-layer CNN combined with LSTM and attention mechanisms for improved accuracy.
Raj et al. (2025)	CNN RNN LSTM GRU	To improve disease detection performance in shrimp, this study introduced a novel disease detection model that combines a hybrid optimization model with an Enhanced Recurrent Capsule Network (ERCN). With a precision of 94.9 %, recall

Table 8 (continued)

Study	DL technique	Objective/Major findings
Saifullah (2025)	LSTM GRU	of 93.5 %, F1-score of 94.6 %, and detection accuracy of 95.2 %, the results verified that the suggested model outperforms traditional RNN, CNN, GRU, and LSTM Networks. To classify chicken egg fertility based on embryonic development seen in egg images, this study investigated the use of LSTM and GRU. It underscored the potential of advanced RNNs in enhancing the efficiency and accuracy of automated farming systems, paving the way for future research to further optimize these models for real-world agricultural applications.
Sharvanthika et al. (2024)	CNN Transformers	This paper introduced a novel approach to mastitis detection by combining CNNs with a Vision Transformer (ViT) model, leveraging the strengths of both architectures. This approach has the potential to revolutionize disease management practices in the dairy industry, leading to better outcomes for farmers and livestock.
Son et al. (2024)	GAN	Using methods such as BlurPool, PixelShuffle, and scSE blocks, this study introduced a novel framework using CycleGAN for improving low-dose CT images in dogs. With high precision and recall, the framework achieved notable improvements in noise reduction, contrast enhancement, and anatomical structure preservation.
Sukkuea and Akkajit (2024)	CNN	This paper demonstrated the application of CNNs in animal science by classifying group-housed pig behaviors, achieving 81.7 % accuracy. It highlighted advancements in real-time monitoring, particularly in identifying stress-related behaviors, thus promoting better animal welfare in agricultural settings.
Tripathy et al. (2025)	CNN	This paper discussed using CNNs for automated animal intrusion detection, enhancing wildlife conservation and agricultural practices by accurately recognizing and sorting animals in images or video streams, thus improving ecosystem protection and security without human intervention.

practices and herd monitoring in large livestock operations. With AI-enabled aerial monitoring systems, grazing patterns can be analyzed, water scarcity can be detected and predators can be identified, enabling farmers to react proactively. These drones work independently and do not require a constant internet connection, which makes them ideal for remote locations where cloud-based solutions are impractical. The Edge AI system also increases the security of the livestock by integrating face recognition and motion tracking systems in the farm cameras, which detect potential threats such as unauthorized access or abnormal behavior of the animals (Mahato and Neethirajan, 2024; Bernabucci et al., 2025).

Beyond environmental management and health monitoring, edge AI is revolutionizing livestock supply chain management. Cutting-edge AI-powered CV systems are used in smart meat processing facilities to evaluate carcass quality, identify contamination, and guarantee adherence to food safety laws. Blockchain-integrated AI platforms and real-time traceability systems built into RFID tags monitor the flow of livestock and animal products from farm to market, lowering fraud and increasing transparency (Pathak et al., 2024; Wang and Li, 2024).

Edge AI has many advantages, but it also has drawbacks, including low processing power in tiny devices, energy shortages, and the

requirement for strong cybersecurity to safeguard locally processed data. Future research must focus on ensuring interoperability across various edge AI systems and devices because widespread adoption will depend on seamless integration. The effectiveness and usability of edge AI solutions in animal and veterinary science will be further improved by ongoing developments in low-power AI chips, federated learning strategies, and decentralized AI models (Pandey, 2024; Gill et al., 2025). As this technology develops, it has the potential to reduce reliance on centralized data-processing infrastructure and to increase the data-drivenness, responsiveness, and sustainability of livestock farming.

3. Challenges and limitations of AI in animal and veterinary science

While AI has made remarkable advancements in animal and veterinary science, its full potential remains constrained by several challenges and limitations. These challenges arise from the inherent complexity of biological systems, resource constraints, data issues, and ethical considerations. Data-related, technological, biological, and system complexity, ethical and social, economic and operational, regulatory and legal, scaling and deployment, and education and training challenges for AI in animal and veterinary science are presented in Tables 9 to 16, respectively.

4. Future perspectives and research directions

The application of AI in veterinary and animal science has the potential to completely transform the discipline by improving accuracy, productivity, and predictive capabilities (Ezanno et al., 2021; Vlaicu et al., 2024). There are several important areas and directions that require more research and development as we proceed.

4.1. Enhanced disease detection and diagnosis

The integration of advanced AI technologies, especially ML and DL, has the potential to revolutionize disease detection and diagnosis in animal and veterinary science (Akinsulie et al., 2024) (Table 17). The ongoing advancement and fine-tuning of ML and DL algorithms are vital for managing the intricacy and vastness of data produced in this sector. For example, CNNs can be utilized to scrutinize medical images, while RNNs can handle sequential data such as animal health records over time (Mienye et al., 2024). Future investigations should focus on transfer learning, a technique that involves enhancing a pre-existing model's performance for related tasks, proving particularly beneficial in situations where labeled datasets are limited. Moreover, XAI plays a significant role, enabling veterinarians to comprehend the reasoning behind AI-driven diagnoses, thereby fostering trust and accelerating the adoption of AI systems in clinical veterinary practice (Linardatos et al.,

2021; Guo and Sycara, 2024).

Effective disease detection often requires the integration of multi-modal data sources. To provide a comprehensive diagnostic picture, imaging (such as X-rays and ultrasounds), genetic information, clinical records, and even environmental factors must be integrated. To improve training sets and increase model robustness, sophisticated data fusion and data augmentation techniques can be used to generate synthetic data. By combining genomic and proteomic data, diseases can be better understood at the molecular level. AI can then find genetic markers associated with specific diseases, creating more accurate diagnostic tools (Li et al., 2023b; Salvi et al., 2024; Xu et al., 2024).

Real-time diagnostic capabilities are essential for timely interventions. By using AI models on edge devices such as sensors and wearables, real-time processing and diagnostics can be performed directly at the point of care, reducing latency and requiring fewer data transfers to central servers. By integrating AI with telemedicine platforms, veterinarians in remote locations can upload medical images or data to cloud-based AI systems for instant diagnostic recommendations (Liu et al., 2024; Uddin and Koo, 2024).

To create accurate and clinically applicable AI models, collaboration between veterinary specialists, AI researchers, data scientists, and other stakeholders is essential. The training of reliable AI models requires the establishment of common databases and standardized methods for data collection, annotation, and sharing. Such cooperation could lead to a significant improvement in the diagnosis and detection of diseases (Akinsulie et al., 2024).

Ethical and practical factors are crucial when developing and implementing AI in veterinary science. It is important to ensure that AI models are impartial, comply with legal requirements, and are applied fairly to different animal species and breeds. It is critical to establish ethical standards and procedures for the verification and approval of AI tools before their implementation in medical environments. Veterinarians' confidence in AI systems is enhanced by training in the effective use and interpretation of these tools (Cohen and Gordon, 2022; Sankar et al., 2024).

The creation of early warning systems and predictive analytics that can predict disease onset before clinical symptoms manifest are examples of future directions in this field. The detection of abnormalities such as tumors, fractures, or infections can be improved by developing AI algorithms for automated image analysis. Real-time health alerts can be obtained by integrating AI with wearable and Internet of Things (IoT) devices for continuous health monitoring. This enables proactive management of animal health. More precise and rapid interventions can occur through personalized health monitoring systems that track the health trends of specific animals over time (Dixon et al., 2024; Maleki Varnosfaderani and Forouzanfar, 2024).

The future of disease detection and diagnosis in animal and veterinary science lies in the integration of advanced AI technologies. By

Table 9
Data-related challenges in AI for animal and veterinary science.

Challenge	Description	Impact	Example	Potential solutions
Data availability	Data is often fragmented, and there is a lack of large, standardized datasets for animal health and welfare	Limited access to comprehensive datasets affects the accuracy of AI models and hinders their generalization across different species and environments	Limited health data on rare diseases in livestock or wild animals	Encourage collaboration across farms, research institutions, and veterinary clinics to create shared databases
Data quality	Animal health data is often noisy, inconsistent, or incomplete, leading to challenges in data preprocessing	Poor-quality data results in inaccurate predictions and lower model performance	Missing data points in animal health records or inconsistent data from different sensor systems	Implement data cleaning techniques and use robust AI models that can handle noisy data
Data annotation	Annotating large datasets for AI applications is time-consuming and requires domain expertise	The lack of sufficient labeled data hampers the training of supervised AI models	Labor-intensive annotation of radiographic images for disease diagnosis in livestock	Utilize semi-supervised learning, crowdsourced annotations, or AI-assisted data labeling
Data sharing and privacy	Concerns about data privacy and ownership prevent data sharing across farms, veterinary clinics, or research institutions	Limited data sharing restricts the development of more generalized AI models	Farmers may be reluctant to share health data with researchers due to privacy concerns	Develop clear data ownership agreements, enforce strong privacy regulations, and establish secure data-sharing platforms

Table 10
Technological challenges in AI for animal and veterinary science.

Challenge	Description	Impact	Example	Potential solutions
Model generalizability	AI models trained on data from one species or environment may not generalize well to others	Models may be ineffective or inaccurate when deployed across different species or farming systems	AI trained to detect lameness in dairy cattle may not apply to sheep or poultry	Use transfer learning and domain adaptation to customize models for different environments
Complex biological systems	Animal health is influenced by numerous interrelated factors (genetics, environment, etc.), making accurate prediction challenging	Models may oversimplify biological complexity, leading to inaccurate predictions	Predicting livestock growth without accounting for all environmental or stress factors	Incorporate multi-factorial models that consider genetics, environment, behavior, and nutrition
Species-specific variations	Differences between species or breeds pose challenges in applying the same AI models across all animals	AI models developed for one species may fail or underperform for others	AI for cattle disease detection may not work for poultry due to different disease patterns	Design species-specific models or create modular AI solutions that can be easily adapted to different species
Behavioral complexity	Animal behavior is context-dependent, and subtle differences in behavior complicate AI analysis	AI models may misinterpret animal behavior, leading to incorrect welfare assessments	Misidentifying stress behaviors in pigs due to varying social dynamics	Use sensor fusion and multi-modal data to better capture behavioral data from diverse contexts
Digitization footprint	AI models require high computational power and increase energy consumption	High resource use limits sustainability, especially in low-resource settings	Use of cloud-based AI solutions requiring significant energy	Develop energy-efficient AI models, optimize computational workflows, and use edge computing

Table 11
Biological and system complexity challenges in AI for animal and veterinary science.

Challenge	Description	Impact	Example	Potential solutions
Multi-factorial influences	Animal health is influenced by a wide range of interconnected factors, such as genetics, diet, environment, and stress	It is difficult to model all these factors simultaneously, leading to incomplete or inaccurate predictions	Predicting milk yield based on genetics without accounting for seasonal variations or feed quality	Develop multi-variable AI models that integrate various types of data (e.g., genomics, feeding patterns, environmental data)
Non-linear dynamics	Biological systems often exhibit non-linear relationships, where small changes can have disproportionate effects	AI models that assume linear relationships may fail to capture critical dynamics in biological systems	A slight change in cattle feed composition may lead to large changes in the growth rate	Use non-linear models like neural networks or ensemble methods to capture complex interactions between factors
Species and breed-specific variability	Different species and breeds within a species exhibit distinct biological characteristics	Models may not work well across diverse animal populations if they are not species- or breed-specific	AI tools developed for Holstein cattle may not apply to other breeds like Jersey or Angus	Customize AI systems for different breeds or species, or use transfer learning to adapt models to new breeds
Behavioral complexity	Animals exhibit diverse and context-dependent behaviors, which are difficult to quantify consistently	Models may fail to detect abnormal behaviors or misinterpret natural behavior as stress or illness	AI for detecting stress behaviors in dairy cows may fail in crowded or noisy barn environments	Improve sensor networks and use behavioral data augmentation to better train AI systems

Table 12
Ethical and social challenges in AI for animal and veterinary science.

Challenge	Description	Impact	Example	Potential solutions
Animal welfare	Data collection for AI systems may cause discomfort, stress, or harm to animals, particularly when using invasive monitoring methods	Animal welfare concerns may arise if AI tools are perceived as intrusive or harmful to animals	The use of wearable sensors on livestock could cause skin irritation or behavioral changes	Design non-invasive AI systems and ensure that animals' physical and emotional welfare is prioritized
Transparency and accountability	AI systems are often seen as "black boxes," making it difficult to understand how decisions are made	Lack of transparency reduces trust in AI, especially in critical decision-making processes like disease diagnosis or treatment	AI might recommend culling sick animals, but the reasoning behind this decision may not be clear	Invest in explainable AI (XAI) methods that provide clear, interpretable reasons for the AI's decisions
Data privacy and ownership	Concerns about the privacy and ownership of data collected on animals and farms	Without clear data ownership and privacy guidelines, stakeholders may hesitate to share data	Farmers may not want to share health data on their livestock with researchers due to fears about data misuse	Implement strong data governance policies, create secure data-sharing platforms, and clarify ownership rights
Liability and accountability	In cases where AI-driven decisions lead to errors (e.g., incorrect disease diagnosis), it's unclear who is responsible	Legal ambiguity in AI applications could lead to disputes over liability in case of harm or financial loss	If an AI system misdiagnoses a disease, it's unclear whether the farmer, veterinarian, or AI developer is liable	Develop clear regulations and legal frameworks around AI accountability in animal health

developing sophisticated algorithms, integrating multimodal data, enabling real-time diagnostics, fostering collaborative efforts, and addressing ethical considerations, AI can significantly increase our ability to detect and diagnose diseases in animals, leading to improved health outcomes, more effective treatments, and better animal welfare (Guo et al., 2023; Karalis, 2024) (Table 17).

4.2. Personalized medicine and treatment plans

Personalized medicine, also known as precision medicine, tailors medical treatment to the individual characteristics of each patient. In the context of veterinary science, this involves customizing health care for animals based on their unique genetic, behavioral, and environmental factors. AI technologies are crucial in advancing personalized medicine by enabling the analysis of large and complex datasets to identify individual-specific health needs and treatment responses (Ho

Table 13
Economic and operational constraints in AI for animal and veterinary science.

Challenge	Description	Impact	Example	Potential solutions
High initial investment	AI systems require significant upfront costs for hardware, software, and installation	Small farms and veterinary clinics may find AI systems unaffordable, limiting their adoption	A disease monitoring system with sensors, cameras, and data analysis tools could cost tens of thousands of dollars	Develop cost-effective AI tools or subscription-based services to reduce the financial burden of small-scale operations
Operational costs and maintenance	Continuous maintenance and updates are required to ensure the proper functioning of AI systems	Ongoing costs related to software updates, sensor calibration, and data storage can be prohibitively expensive for smaller operations	A system for monitoring animal health that requires regular updates and recalibration of sensors	Offer cloud-based AI solutions that reduce on-site maintenance costs or integrate remote monitoring services
Return on Investment (ROI)	Small-scale operations may struggle to see the ROI from AI adoption, particularly in the short-term	Without clear and immediate ROI, small farmers may hesitate to adopt AI technologies	AI-based feeding systems may not immediately show improvements in yield or efficiency	Create pilot programs to demonstrate clear ROI, and develop affordable, scalable solutions that offer gradual benefits
Scalability	AI solutions developed for small farms may struggle to scale to larger operations, or vice versa	AI solutions may not be adaptable or cost-effective for farms of different sizes or types	A system that works well in a small dairy farm may not scale to a large industrial farm	Design modular, scalable AI systems that can be customized for different sizes and types of operations

Table 14
Regulatory and legal challenges in AI for animal and veterinary science.

Challenge	Description	Impact	Example	Potential solutions
Lack of standardized regulations	There are no universally accepted standards for AI applications in animal health	The absence of regulatory consistency leads to uncertainty about the safety and effectiveness of AI tools	An AI system for disease prediction may not undergo the same rigorous validation as medical devices	Establish international regulatory bodies and create standardized frameworks for AI tools in veterinary science
Data privacy and ownership	The lack of clear regulations around the privacy and ownership of animal health data creates confusion	Farmers and veterinarians may be hesitant to share sensitive data, limiting the development of comprehensive AI models	Concerns about data ownership when farms share health data with researchers or tech companies	Implement clear data privacy laws and data-sharing agreements to protect stakeholders' rights
Liability and accountability	It is unclear who is responsible if an AI system makes an incorrect decision leading to harm	The lack of clear legal accountability can hinder the adoption of AI systems due to fears of litigation	If an AI-driven diagnostic tool misdiagnoses a condition, there's ambiguity about who is legally liable	Develop legal frameworks that define accountability for AI decisions in veterinary practice
Regulatory approval processes	AI tools may face regulatory hurdles that delay their approval for widespread use	Slow or unclear regulatory approval processes can stifle innovation and delay market entry	AI systems for disease detection may face long approval timelines, delaying their use in clinical practice	Streamline approval processes for AI tools in veterinary science and create faster pathways for deployment

Table 15
Scaling and deployment challenges in AI for animal and veterinary science.

Challenge	Description	Impact	Example	Potential solutions
Data integration	Data from various sources (e.g., farms, clinics, sensors) may not be compatible	Inconsistent data from different sources can make it difficult to train and deploy AI systems effectively	Integration of farm-level data (e. g., GPS collars, sensors) with veterinary health records	Standardize data formats and establish common data exchange protocols to improve integration
Infrastructure limitations	AI systems require high computational power, data storage, and reliable internet connections, which may not be available in rural areas	Lack of infrastructure limits the ability of small-scale farms or rural clinics to deploy AI tools	Farms without access to cloud computing or high-speed internet may struggle to implement AI solutions	Promote the use of edge computing for local processing, and provide affordable, scalable cloud solutions
Customization and adaptability	AI systems need to be adapted to specific animal species, breeds, or environments	High costs and time requirements for customization limit scalability	An AI system for poultry health may need to be entirely redesigned for swine due to different behaviors and diseases	Develop modular AI systems that can be easily customized for different species, breeds, or environmental factors
Cost of training and retraining	AI models require significant investment in data annotation, model training, and updates	The high costs of training and retraining models may hinder adoption, particularly for small-scale operations	Continual updates and retraining of disease prediction models using new datasets	Use transfer learning and semi-supervised learning to reduce training costs and reliance on large labeled datasets
Sustainability of AI in farming	Deployment of AI technologies can have significant environmental and computational footprints	High computing demands in precision livestock systems increase carbon emissions	AI-powered disease monitoring systems that require constant data processing	Promote energy-efficient AI, edge computing, and sustainable infrastructure development

et al., 2020; Marques et al., 2024b) (Table 18).

AI can analyze genomic data to identify genetic variations associated with specific diseases or health conditions, predicting disease susceptibility and tailoring preventive measures or treatments based on an animal's genetic makeup. By integrating proteomic and metabolomic data, AI provides insights into the molecular mechanisms underlying diseases, allowing for a comprehensive understanding of an animal's health and enabling the development of targeted therapies. Moreover, AI can

analyze epigenetic modifications (changes in gene expression without altering the DNA sequence) to understand how environmental factors influence gene expression, designing personalized treatment plans that consider both genetic and environmental influences (Sharma et al., 2024; Vilhekar and Rawekar, 2024).

AI can develop predictive models that forecast the progression of diseases based on an animal's health history, genetic data, and current health status. These models identify early warning signs and suggest

Table 16
Educational and training gaps in AI for animal and veterinary science.

Challenge	Description	Impact	Example	Potential solutions
Lack of AI knowledge	Many veterinarians and animal care professionals lack basic knowledge of AI and data science	This limits the adoption and effective use of AI systems in clinical and farm settings	Veterinarians may hesitate to adopt AI-driven diagnostic tools because they don't understand how the models work	Develop AI literacy programs and integrate data science into veterinary training curricula
Shortage of AI experts in animal science	There is a shortage of AI specialists with domain knowledge in animal health and veterinary medicine	The lack of cross-disciplinary expertise hinders the development and deployment of AI solutions tailored to animal science	Few AI experts understand both the genomics of livestock and the technical aspects of deep learning	Create interdisciplinary education programs that combine AI with animal health training, and foster collaboration between AI engineers and veterinarians
Hands-on training	Limited opportunities for veterinarians and farmers to gain practical experience with AI tools	Lack of hands-on experience leads to underuse or misuse of AI technologies in the field	Veterinarians may not feel comfortable using AI-driven diagnostic platforms without proper training	Implement pilot programs and simulation-based training for veterinarians and farmers to get hands-on experience with AI applications
Lack of resources in rural areas	Small-scale farms and rural veterinary clinics may lack the resources to adopt AI technologies	This results in a digital divide between large industrial farms and small rural operations	Smaller farms may not have access to the necessary infrastructure or budget to deploy AI-based animal health monitoring systems	Develop affordable AI solutions and provide financial incentives or subsidies for small-scale farmers to adopt AI tools

timely interventions to prevent disease progression. By analyzing historical treatment data, AI can predict how an individual animal will respond to different treatments, allowing veterinarians to select the most effective treatment for each animal with the lowest risk of adverse effects. Furthermore, AI can assess the risk of developing certain diseases based on various factors, including genetics, lifestyle, and environmental exposures, allowing for proactive measures to reduce risk and improve overall health outcomes (Guo et al., 2023; Dixon et al., 2024; Maleki Varnosfaderani and Forouzanfar, 2024).

AI-powered systems can continuously monitor an animal's health using data from wearable devices, sensors, and clinical records. These systems detect changes in health status in real-time and adjust treatment plans accordingly. AI can create dynamic treatment plans that adapt to the changing health condition of the animal and are updated based on new information, such as changes in vital signs or response to treatment, to ensure optimum care. Moreover, AI can determine the precise dosage of medications needed for each animal, considering factors such as weight, age, genetic makeup, and current health status, reducing the risk of over- or under-dosing and enhancing treatment efficacy (Shajari et al., 2023; Michielon et al., 2024).

It is essential to ensure the privacy and security of animal health data and that AI systems comply with ethical guidelines and regulations for the protection of sensitive information. Personalized medicine should be accessible to all animals, regardless of species, breed, or socioeconomic status, with efforts made to develop affordable and widely available AI-based solutions. AI systems should be transparent in their decision-making processes, allowing veterinarians to understand and explain the rationale behind AI-generated recommendations to animal owners (Harishbhai Tilala et al., 2024).

The future of personalized veterinary medicine includes the integration of AI systems with electronic health records (EHRs) to provide comprehensive and up-to-date information on an animal's health status, enhancing the accuracy of personalized treatment plans. The development of AI-driven clinical decision support systems (CDSS) can assist veterinarians in making informed decisions by providing real-time analysis of clinical data, suggesting potential diagnoses, and recommending personalized treatment options. Collaborative research between veterinarians, researchers, AI developers, and data scientists can advance the field of personalized veterinary medicine, leading to the development of more robust and accurate AI models. Additionally, providing education and training for veterinarians on the use of AI tools in personalized medicine ensures that they can effectively utilize AI systems to improve patient care (Alanazi, 2023; Akinsulie et al., 2024; Hirani et al., 2024).

4.3. Precision agriculture and livestock management

By using cutting-edge technologies to monitor and control livestock productivity, behavior, and health, PLF aims to optimize farming practices (Tables 19 to 20). In this respect, AI is essential, as it provides data-driven insights that improve operational efficiency and decision-making (Vlaicu et al., 2024).

Using sensors, cameras, and other IoT devices, AI-powered systems can automate animal health monitoring. These systems enable real-time health assessments by continuously collecting information on vital signs, physical activity, and behavioral patterns. Vital signs such as body temperature, respiratory rate, and heart rate can be monitored through wearable technology with built-in sensors. AI systems examine this data to find deviations from typical areas that could indicate medical problems. Early disease detection and timely intervention are made possible through continuous monitoring (Rajak et al., 2023; Shahab et al., 2024).

Cameras are used by AI-powered CV systems to track animal behavior. AI can detect behavioral problems that could indicate stress or health problems by studying social interactions, eating habits, and exercise patterns. For example, a change in posture or gait can be a sign of lameness, while reduced eating activity can be a sign of illness. Air quality, humidity, and temperature can be monitored by sensors in animal stables. AI examines this information to maintain ideal living conditions, reduce stress, and improve health. For example, breathing problems can be avoided by adjusting ventilation systems in response to real-time air quality data (Okinda et al., 2020; Neethirajan, 2023; Saud et al., 2024).

By evaluating information about animal growth, health, and nutritional needs, AI can improve feeding practices. This ensures that every animal gets the right type and quantity of food and promotes healthier growth. AI systems can assess the nutritional value of different feed types and determine the best feed for a variety of animals. To produce a balanced and needs-based diet, the protein, carbohydrate, fat, vitamin, and mineral content of the feed should be assessed. Automated feeding systems powered by AI can deliver feed at exactly the right time, reducing waste and ensuring consistent nutrition. These systems maximize feed efficiency and increase productivity by adjusting feed schedules and quantities in response to real-time data on animal growth and activity. To identify patterns and trends in feed consumption and conversion rates, AI can analyze data. AI can suggest changes to increase feed efficiency, reduce costs, and improve animal performance by understanding how different feed types affect growth and health (Akintan et al., 2024; Arshad et al., 2024; Can, 2024; Zhang et al., 2024a).

By examining genetic information and medical records to determine the most suitable breeding pairs, AI improves breeding programs. As a result, farm animals have higher genetic quality, making them healthier and more productive. Genetic markers linked to desirable attributes

Table 17
Development of advanced AI models for animal and veterinary science.

Aspect	Description	Applications	Future research directions
Improving model accuracy and generalization	Creating models that adapt to diverse species, breeds, and environments while maintaining predictive accuracy	Cross-species disease prediction, global livestock monitoring	Develop <i>meta</i> -learning and transfer learning techniques to adapt models for new species or tasks with minimal retraining
Hybrid AI models	Combining rule-based systems with machine learning to improve performance, especially with sparse or noisy data	Disease diagnosis, behavioral monitoring	Create explainable hybrid models that integrate domain-specific knowledge with AI insights
Context-aware AI	Incorporating environmental and situational data (e.g., climate, feed quality) into predictions for better insights	Resource management, seasonal disease forecasting	Develop adaptive AI systems using multi-modal data fusion and real-time dynamic learning frameworks
Advancements in deep learning	Enhancing deep learning models to process high-dimensional and unstructured data like images, videos, and genomic sequences	Image-based diagnostics, behavioral analysis, genomic data interpretation	Focus on lightweight models, explainable deep learning, and transformer architectures for long-term dependency analysis
Multi-agent AI systems	AI systems where multiple agents collaborate to solve complex problems by integrating data from various sources	Livestock health monitoring, epidemiological modeling for disease spread	Design scalable multi-agent systems with efficient communication protocols for distributed farm-level or clinic-level use
Leveraging emerging technologies	Using advanced tools like quantum computing, edge AI, and federated learning for faster, scalable, and privacy-preserving AI models	Precision livestock farming, collaborative disease surveillance	Explore quantum-inspired algorithms and federated learning for decentralized data-sharing across farms and clinics
Automation in AI development (AutoML)	Automating AI model creation to lower entry barriers and enable non-experts to build and deploy predictive models	Small-scale farm disease monitoring, breeding optimization	Expand AutoML frameworks with templates tailored to common tasks in animal science, such as feeding or health monitoring

such as growth rate, productivity, and disease resistance can be analyzed by AI algorithms. By selecting breeding pairs with desirable characteristics, AI helps improve the overall genetic quality of livestock populations (Pedrosa et al., 2024). AI is capable of building models that predict the outcomes of breeding programs, including the likelihood that offspring will inherit certain traits. These models help farmers make informed decisions by providing accurate predictions based on genetic information, environmental factors, and historical breeding data (Vilhekar and Rawekar, 2024). By predicting ideal breeding times based

on hormonal data and other physiological indicators, AI can improve reproductive management. This reduces the time and resources required for reproductive management and increases the success rate of breeding programs (Chafai et al., 2023).

AI optimizes the use of resources and minimizes waste, helping make animal husbandry less environmentally harmful. This reduces the agricultural carbon footprint and encourages sustainable agricultural practices. AI systems can monitor and control resource consumption, such as feed, energy, and water. These systems help reduce waste and increase efficiency by optimizing resource use. For example, AI can evaluate irrigation data to ensure optimal water use, avoid overwatering, and conserve water supplies. AI is capable of analyzing waste production data and creating effective waste management plans. This includes composting or recycling manure to produce bioenergy, which reduces the impact on the environment and gives farmers access to new sources of income. Crop rotation, conservation tillage, and integrated pest management are examples of sustainable farming techniques that AI can control. These practices support the long-term sustainability of agriculture by maintaining soil health, reducing chemical use, and promoting biodiversity (Ahuja and Mehra, 2023; Onyeaka et al., 2023; Shahab et al., 2024).

Integrating cutting-edge AI technologies and creative solutions is critical to the future of precision agriculture and livestock farming. Farmers can obtain real-time information by creating integrated platforms that integrate AI analytics with IoT devices. These systems can optimize farm operations and improve decision-making by monitoring a variety of parameters, including environmental conditions, feed efficiency, and animal health. The traceability of animal products can be improved in practice by using blockchain technology. AI ensures accountability and transparency across the supply chain by tracking and recording data on animal health, breeding, and feeding practices. Customers can use this data to confirm the origin and quality of goods (Karunathilake et al., 2023; Duguma and Bai, 2024; Padhiary et al., 2024).

Farmers can receive practical advice based on data analysis by developing AI-powered decision support systems. By helping farmers make informed decisions regarding disease prevention, resource allocation, and herd management, these systems can increase the overall productivity and efficiency of farms. Proactive steps to curb the spread of disease and protect the health of livestock are enabled by developing AI models that can predict disease outbreaks based on past data, environmental factors, and animal health indicators. Farmers can act promptly by informing them of possible dangers through early warning systems. Feeding, milking, and cleaning are examples of labor-intensive tasks that can be automated by combining AI and robotics. Using robotic systems to complete these tasks precisely and consistently can reduce farmer workloads and increase operational efficiency. For example, robotic milking machines can guarantee consistent milk quality and optimize milking schedules (Cogato et al., 2021; Elbasi et al., 2024; Mana et al., 2024).

By constantly monitoring behavior and health, providing ideal living conditions, and detecting health problems early, AI can help improve animal welfare. This makes the animals happier and healthier, which leads to an increase in productivity and product quality (Dawkins, 2025). AI can analyze crop performance, soil conditions, and weather patterns to help farmers develop climate-resilient practices. This makes it possible to take measures that make animal husbandry more resilient to climate change and ensure sustainable production despite environmental problems (Neethirajan et al., 2024; Araújo et al., 2025).

Using AI to improve productivity, sustainability, and efficiency, precision agriculture and livestock management are revolutionary concepts in farming. The welfare and productivity of livestock can be greatly increased by AI-driven solutions that automate health monitoring, optimize feeding procedures, improve breeding programs, and lessen their negative effects on the environment. The future of this sector depends on the continued development and integration of cutting-edge

Table 18

AI in precision medicine for veterinary care.

Aspect	Description	Applications	Challenges	Future directions
Genomic-based treatment	Analyzes genomic data to tailor treatments for individual animals based on genetic predispositions	Identifies genetic markers for disease resistance, guides breeding programs, and customizes treatments	Requires extensive genomic data, which may be unavailable for rare species or breeds	Establish global genomic repositories for veterinary applications
Drug dosage optimization	Predicts the most effective drug and dosage based on species, breed, and metabolism	Reduces side effects and enhances treatment efficacy by personalizing medication protocols	High computational demand for real-time optimization in clinical settings	Develop lightweight AI algorithms for use in small clinics
Disease risk prediction	Uses health records and environmental data to forecast disease susceptibility and guide preventive care	Enables proactive interventions, reducing disease incidence and improving herd health	Limited integration of environmental and genetic data in predictive models	Build AI systems that seamlessly integrate multi-source data for disease risk analysis
Real-time health monitoring	Wearables track vital signs and detect early signs of illness, enabling timely interventions	Prevents disease progression and supports recovery monitoring post-treatment	Expensive devices may limit adoption in small-scale farms or resource-constrained regions	Design affordable wearable devices with extended battery life and wireless connectivity
Predictive analytics for chronic diseases	Forecasts the progression of chronic conditions and recommends management strategies	Improves long-term health outcomes by tailoring care plans for conditions like diabetes or arthritis	Requires high-quality longitudinal health data, which is often incomplete	Encourage the use of electronic health records to build robust datasets for chronic disease analysis
Nutrition personalization	Recommends diets tailored to an animal's metabolic profile and health needs	Optimizes feed efficiency and supports recovery from illness or injury	Data collection on metabolic and dietary factors can be time-intensive and costly	Integrate AI tools with feed analysis systems for real-time diet recommendations
Veterinary pathology	AI analyzes tissue samples and imaging data for accurate diagnosis and prognosis	Enhances diagnostic precision for conditions like cancer or infectious diseases	Lack of standardized imaging and pathology datasets for AI training	Collaborate with pathology labs to create comprehensive datasets for veterinary AI tools
Data availability and quality	Comprehensive datasets, including genomic, environmental, and health records, are critical for precision medicine	Enables accurate predictions and personalized care for individual animals	Fragmented or incomplete data limits AI model training and precision	Promote standardized data collection protocols across veterinary practices
Cost-effective tools	Affordable AI-powered tools make precision medicine accessible to smaller veterinary clinics and farmers	Ensures equitable access to advanced diagnostic and treatment capabilities	High costs of current AI technologies create barriers to widespread adoption	Develop modular AI tools with scalable features to reduce costs for small clinics
Integrated health platforms	Combines genomic, health, and environmental data into a single AI-driven platform for holistic analysis	Improves diagnostic accuracy and treatment planning by analyzing interconnected factors	Integration challenges due to varying data formats and collection methods	Build interoperable systems with standardized APIs for seamless data sharing
Ethical AI development	Ensures AI systems are transparent, unbiased, and respect data privacy and ownership	Builds trust in AI-driven precision medicine among veterinarians and pet owners	Lack of clear ethical guidelines for veterinary AI applications	Establish ethical frameworks specifically tailored for AI use in veterinary medicine

Table 19

AI for sustainable agriculture in livestock farming.

Aspect	Description	Applications	Challenges	Future directions
Precision manure management	Optimizes manure use by analyzing nutrient content and application timing to reduce environmental runoff	Enhances soil health, reduces nutrient pollution, and improves crop yield	Requires data from soil tests and manure composition, which may not be readily available	Integrate soil and manure sensors for automated analysis and real-time recommendations
Monitoring greenhouse gas (GHG) emissions	Uses AI-enabled sensors to measure and analyze methane and ammonia emissions from livestock operations	Identifies high-emission areas and optimizes feeding practices to reduce emissions	Sensor costs and variability in emission measurements across farming systems	Develop low-cost, portable sensors and AI models that normalize variability in emission data
Water management and conservation	Optimizes water use with smart irrigation systems and AI algorithms that analyze moisture levels and weather patterns	Prevents water wastage, ensures efficient irrigation, and reduces stress on water resources	Requires robust infrastructure and connectivity, which may be unavailable in rural areas	Use edge computing and solar-powered IoT devices for water monitoring in off-grid locations
Optimizing feed efficiency	Analyzes feed quality and animal growth data to design optimal rations, reducing feed waste and cost	Improves animal growth rates, reduces nutrient excretion, and enhances feed conversion efficiency	Data on feed composition and animal performance may not be available in small farms	Use AI tools that integrate historical feed data with live monitoring to provide personalized feed recommendations
Promoting biodiversity	Supports integrated farming systems that preserve natural habitats and enhance ecosystem services	Enhances pollination, improves soil quality, and increases resilience to environmental changes	Balancing productivity with biodiversity goals can be challenging in intensive farming systems	Develop AI models that assess trade-offs and recommend biodiversity-friendly farming practices
Land use optimization	Evaluates land suitability for grazing and crop cultivation based on soil quality, climate, and biodiversity data	Prevents overgrazing, promotes sustainable crop-livestock integration, and conserves natural resources	Access to detailed land use and environmental data is often limited	Integrate satellite imagery and AI for large-scale land-use assessment and monitoring
Supply chain optimization	AI predicts demand, manages inventory, and enhances traceability in the livestock product supply chain	Reduces food waste, improves energy efficiency, and aligns production with market demand	Requires coordination across the supply chain and access to real-time market data	Implement blockchain-enabled AI systems for transparent and efficient supply chain management
Lifecycle analysis models	AI evaluates the environmental impact of farming practices across the entire production lifecycle	Identifies high-impact areas and recommends mitigation strategies to lower carbon footprints	Complex models require diverse datasets that are often unavailable or incomplete	Create standardized frameworks for lifecycle data collection and analysis, supported by AI
AI for climate adaptation	Predicts climate-related risks like heat stress or droughts and recommends adaptive measures	Ensures livestock health and productivity under changing climate conditions	Requires region-specific climate data and long-term weather forecasts	Design AI models tailored for regional climates and integrated with meteorological data sources

Table 20
Robotics and automation in animal and veterinary science.

Aspect	Description	Applications	Challenges	Future directions
Automated milking systems (AMS)	Robots milk cows efficiently, analyze milk quality, and provide real-time data on udder health	Reduces labor, enhances productivity, and provides early mastitis detection	High initial cost limits adoption for small farms	Develop cost-effective AMS systems with scalable options for small-scale farms
Robotic feeders	Delivers precise rations based on individual or group needs to optimize feed efficiency and reduce waste	Improves growth rates, reduces nutrient wastage, and enables personalized feeding	Requires integration with feed quality and animal performance data	Enhance feeders with real-time AI models that adjust rations dynamically
Cleaning and waste management robots	Automatically clean barns and manage manure, reducing labor and improving hygiene	Prevents disease, maintains cleanliness, and reduces manual labor	Regular maintenance and adaptability to different barn layouts	Develop self-learning robots capable of adapting to various environments
Health monitoring robots	Track vital signs, movement patterns, and body condition for early health issue detection	Provides early warning for diseases and monitors recovery post-intervention	High dependency on robust sensor data for accuracy	Integrate advanced sensors and AI for multi-parameter health analysis
Robotic surgery and assistance	Assist veterinarians in complex surgical procedures, improving precision and reducing recovery times	Enhances success rates in surgeries like orthopedic or soft tissue procedures	Expensive and requires specialized operator training	Design user-friendly robotic surgical systems with remote operation capabilities
Livestock sorting and handling	Automates animal sorting by health status, weight, or behavior, reducing stress during handling processes	Improves efficiency in large-scale farms and minimizes injuries to animals	Requires significant calibration for diverse animal sizes and species	Build adaptive systems capable of handling multiple species and housing layouts
Drone-assisted monitoring	Uses drones equipped with cameras and sensors to monitor large herds and grazing lands	Identifies sick or isolated animals and maps pasture health remotely	Limited battery life and susceptibility to weather conditions	Develop drones with extended battery life and rugged designs for outdoor use
Multi-function robots	Robots perform various tasks like feeding, cleaning, and health monitoring to maximize efficiency	Streamlines operations, reduces labor costs, and improves farm productivity	Complexity in designing multi-tasking robots that maintain performance across tasks	Create modular robots that allow farmers to add or remove functions as needed
Energy and resource optimization	Ensures robotic systems operate efficiently, reducing power and resource consumption for sustainable use	Enhances sustainability by minimizing energy costs and environmental impact	Dependence on reliable energy sources in off-grid areas	Design renewable energy-powered robots for continuous operation in rural regions
AI-driven autonomy	Robots use AI to make decisions, such as identifying health issues or adjusting feed delivery schedules	Reduces the need for human supervision and allows real-time responses to environmental or health changes	Requires robust AI algorithms and comprehensive training datasets	Invest in advanced AI models capable of multi-tasking and decision-making with limited human input

technologies to create a sustainable and efficient agricultural ecosystem. As AI technologies advance, they will become more crucial in tackling the issues facing the agriculture sector, guaranteeing food security, and advancing sustainable development (Dayoub et al., 2024; Mana et al., 2024; Rosati, 2025).

4.4. AI in animal welfare and ethics

The field of integrating AI with animal welfare and ethics is rapidly evolving and seeks to improve animal welfare while addressing moral dilemmas. AI technologies that offer automated care systems, predictive analytics, and sophisticated monitoring have the potential to

Table 21
AI for behavioral and welfare assessment in animal and veterinary science.

Aspect	Description	Applications	Challenges	Future directions
Behavioral monitoring	Tracks eating, resting, walking, and social behaviors to identify normal and abnormal patterns	Detects lameness, reduced feed intake, or social isolation in livestock	Requires high-quality video or sensor data for accurate analysis	Develop species-specific computer vision models for improved behavioral tracking
Stress and pain detection	Uses facial expressions, posture, and vocalization patterns to assess stress and pain levels in animals	Early detection of stress-related issues like heat stress or transport-related discomfort	Data variability due to environmental factors like lighting or background noise	Enhance AI models with noise-resilient algorithms for reliable stress detection
Environmental monitoring	Integrates data from sensors to ensure optimal temperature, humidity, and air quality for animal comfort	Prevents heat stress, ammonia-related respiratory issues, and other environmental hazards	Maintenance and cost of environmental sensors in large-scale operations	Use IoT-enabled sensors with self-maintaining capabilities for efficient monitoring
Automated welfare scoring	Assigns objective scores for metrics like body condition or gait using AI-driven image analysis	Ensures consistent welfare assessments without human subjectivity	Difficult to scale due to species-specific scoring requirements	Standardize scoring systems across species and integrate them with cloud-based analytics
Aggression and social dynamics	Tracks interactions among animals to detect aggressive behavior or social hierarchies in group housing	Reduces injuries and stress in crowded environments like poultry farms or feedlots	Computationally intensive, especially in dynamic group settings	Optimize AI models for real-time social behavior analysis in large animal groups
Data quality and availability	High-quality behavioral and environmental data are critical for accurate AI model training	Facilitates robust behavior monitoring and welfare assessments	Limited datasets due to inconsistent data collection protocols	Create open-access databases with standardized behavioral data across species
Hardware limitations	Cameras and sensors may face issues like maintenance needs and data loss in extreme environments	Enables continuous monitoring of welfare parameters	Expensive hardware may limit adoption in resource-constrained settings	Develop low-cost, durable hardware solutions for rural and small-scale farms
AI-powered decision support systems	Combines behavior and environmental data to provide actionable welfare recommendations	Guides farmers in interventions like feeding adjustments or stress management	Users may struggle to interpret AI recommendations without sufficient training	Build intuitive interfaces that simplify recommendations for farmers and veterinarians
Wearable AI devices	Real-time tracking of physiological metrics such as heart rate, body temperature, and movement	Identifies early signs of disease or stress, improving response times	May cause discomfort or behavioral changes in some animals	Design lightweight, non-invasive wearables with enhanced battery life and data connectivity

significantly improve animal welfare (Table 21). To ensure that these technologies benefit animals without inadvertently harming them, it is important to consider their ethical implications (Zhang et al., 2024a).

By enabling automated care systems, predictive analytics, and continuous, in-depth monitoring, AI technologies have the potential to completely transform animal welfare. For example, AI-controlled systems can continuously monitor the behavior and health of animals and detect early signs of illness or stress. Vital signs such as body temperature, respiratory rate, and heart rate can be tracked through wearable technology with built-in sensors. AI systems examine this data to find deviations from typical ranges that could indicate health problems. Better health outcomes and timely interventions are enabled by early disease detection enabled by continuous monitoring (Michielon et al., 2024; Dawkins, 2025; Rosati, 2025).

To improve animal welfare, AI can also detect signs of stress, such as excessive vocalizations or agitation, and alert nursing staff to take appropriate action (Okinda et al., 2020; Neethirajan et al., 2024; Saoud et al., 2024).

Environmental factors such as temperature, humidity, air quality, and lighting can be monitored through sensors installed in animal stalls. AI examines this information to maintain ideal living conditions, reduce stress, and improve health. AI can also detect environmental changes such as high temperatures or poor air quality that could affect animal welfare and send alerts to take immediate action (Neethirajan, 2024).

By automating routine care tasks such as cleaning, feeding, and monitoring, AI can reduce the workload on human caregivers and ensure consistent care for animals. For example, robotic systems can provide enrichment activities, clean enclosures, and dispense medications. To ensure animals receive the right level of care and attention, automated systems can also adjust care plans in response to real-time data (Dilaver and Dilaver, 2024).

While AI has the potential to improve animal welfare, ethical issues must be considered to ensure these technologies are used wisely. Large amounts of data on animal behavior and health are collected and analyzed by AI systems. This data must be protected to prevent misuse or illegal access. Maintaining the trust of stakeholders, carers and pet owners depends on data security and privacy. To protect sensitive data and ensure responsible data handling, AI systems should comply with ethical standards and laws (Coghlan and Parker, 2024; Rosati, 2025).

Veterinarians and animal breeders should be able to understand and justify the reasons for AI-generated recommendations, as AI algorithms should be open and honest when making decisions. Transparency promotes trust and ensures the ethical and responsible use of AI systems. AI developers should provide clear documentation of algorithm operation, data sources, and possible biases (Akinsulie et al., 2024).

When designing AI systems, it is important to avoid biases that could lead to unfair treatment of different animal species or breeds. Developers should ensure that algorithms are tested for fairness and that training data is representative of different animal populations. Preventing discrimination and providing equitable care depends on eliminating bias in AI models (Ferrara, 2024).

The possibility of unforeseen harm from AI technologies must be taken into account. For example, an overreliance on automation in livestock farming can lead to a disregard for the need for compassionate care and the human-animal relationship. Instead of replacing human care, AI systems should be developed to improve it. The importance of maintaining the emotional and social well-being of animals should be highlighted in ethical guidelines (Vlaicu et al., 2024; Rosati, 2025).

It is important to create comprehensive ethical standards and guidelines for the use of AI in animal welfare. These rules should cover issues such as fairness, transparency, data protection, and possible impacts on animal welfare. To set and maintain these ethical standards, collaboration between researchers, veterinarians, animal welfare groups, and AI developers is critical (van Hilten et al., 2025).

Creating moral standards, cutting-edge technology, and collaborative initiatives to ensure the ethical and beneficial use of AI are key to

the future of AI in animal welfare and ethics. Animal welfare can be significantly improved by creating AI-powered decision support systems that provide practical advice on animal care. To ensure each animal receives the best possible care, these systems can evaluate data on behavior, health, and environmental factors to create individualized care plans. Veterinary professionals and caregivers can use decision-support systems to make informed decisions regarding management practices, nutrition, and medical treatments (Mana et al., 2024; Neethirajan, 2024).

By examining past data, environmental factors, and animal health indicators, AI can create predictive models for disease outbreaks. The ability of these models to identify trends and early warning indicators of disease outbreaks enables proactive measures to contain the spread of disease and protect animal health. Moreover, vaccination schedules, biosecurity precautions, and quarantine procedures can be improved through the use of predictive analytics (Guo et al., 2023; Zhao et al., 2024).

Blockchain technology can be used to ensure supply chain transparency and improve the traceability of animal products. AI ensures a secure and impenetrable record by tracking and recording information about animal health, breeding, and feeding practices. Customers can use this information to confirm the origin and quality of animal products, promote moral behavior in animal husbandry, and increase consumer trust. In addition, blockchain technology can help with certification and regulatory compliance (Ellahi et al., 2023; Patel et al., 2023).

By tracking poaching activities, monitoring endangered species, and assessing habitat conditions, AI can significantly contribute to wildlife conservation. AI-powered cameras on drones enable aerial photography, wildlife identification, and movement tracking. To identify and monitor wildlife populations, AI can also examine data from camera traps, satellite images, and acoustic recordings. AI can support conservation efforts and help protect biodiversity by providing insightful information (Fergus et al., 2024; Reynolds et al., 2024).

By tracking the condition and behavior of captive animals, AI can improve animal welfare in zoos and aquariums. AI-powered systems can detect symptoms of stress or illness by analyzing data from sensors, cameras, and medical records. Interventions and changes to the animals' environment, occupational activities, and care plans can be guided using this information. AI can also facilitate the performance of studies on animal health and behavior in captivity and improve management techniques (Congdon et al., 2022; Liptovszky, 2024).

It is critical to educate and train veterinarians, animal care workers, and AI developers on the moral application of AI in animal welfare. This ensures that everyone involved is aware of the moral issue and can use AI technologies to improve animal care. Topics such as bias reduction, algorithm transparency, privacy, and compassionate care practices can be covered in training programs. The creation of comprehensive training programs can be supported by collaboration between academic institutions, trade associations, and business partners (Coghlan and Quinn, 2024).

AI offers automated care systems, predictive analytics, and sophisticated monitoring, all of which have the potential to significantly improve animal welfare. To ensure responsible use of AI technologies, it is imperative to address ethical issues such as data protection, transparency, fairness, and unintentional harm. Promoting animal welfare and ensuring the ethical application of AI in this area requires teamwork, the creation of moral standards, and continued technological advances. We can build a future in which technologies improve the care, health, and welfare of animals while maintaining the highest ethical standards, by responsibly using AI (Olawade et al., 2024; Rosati, 2025).

4.5. AI-assisted veterinary education and training

The use of AI in veterinary education and training promises great potential for improving clinical skills, learning experiences, and continuous professional development of veterinarians. AI technologies

can help veterinary professionals and students by providing real-time decision support, personalized training, and immersive, interactive learning resources (Table 22). These developments have the potential to revolutionize the acquisition and application of veterinary knowledge (Akinsulie et al., 2024).

The teaching and application of veterinary knowledge could be completely transformed by the use of AI in veterinary education and training. Virtual reality (VR) environments and AI-powered simulation platforms can provide realistic scenarios for veterinary students to hone their skills. These resources allow students to perform risk-free, controlled virtual surgeries, diagnose diseases, and create treatment plans. Complex concepts and processes are better understood and retained through interactive learning. AI can provide personalized learning materials and feedback by analyzing each learner's learning preferences, strengths, and deficits. Lessons can be tailored to the needs of individual students through adaptive learning platforms, ensuring they receive the right level of support and challenge. This individualized approach allows students to learn more effectively and progress at their own pace. CDSS with AI capabilities can help veterinarians make smart decisions in clinical settings. These systems provide real-time diagnosis and treatment recommendations by analyzing patient data, medical records, and research results. Better patient care can be achieved through more accurate and effective clinical decision-making through CDSS (Riddle et al., 2024; Sousa and Flay, 2024).

For veterinarians, AI may support continuous professional development and lifelong learning. Veterinary practitioners should be kept informed of the latest developments in veterinary medicine through webinars, online courses, and AI-based learning platforms. To keep veterinarians up to date on new information and skills, AI can also monitor each learner's progress and suggest appropriate educational materials. Numerous aspects of veterinary education and training can benefit from the use of AI technologies. Students can practice their diagnostic and surgical skills using AI-powered virtual simulations that mimic real-world clinical scenarios. These simulations provide a safe environment in which students can grow from their mistakes without endangering real animals. Cutting-edge VR and augmented reality (AR) technologies can create engaging training scenarios that enhance

experiential learning. For example, VR simulations can recreate complicated surgical procedures, allowing students to hone their skills in a virtual environment before applying them to real animals (Venkatesan et al., 2021; Akinsulie et al., 2024; Chu et al., 2024; Esseme et al., 2025).

Based on actual patient data, AI can create interactive case studies that allow students to apply what they have learned to actual circumstances. These case studies can cover everything from common clinical conditions to unusual illnesses, and develop students' ability to think critically and solve problems. AI can also adjust the difficulty level of case studies according to student progress, ensuring the best possible learning environment. The assessment of students' performance in tests and practice can be automated by AI. Intelligent tutoring systems can provide immediate feedback on clinical skills such as surgical or diagnostic techniques. This feedback allows students to improve their skills and identify growth potential. For example, AI can evaluate a student's performance during a mock surgery and provide detailed comments on their accuracy, technique, and adherence to best practices (Alowais et al., 2023; Bekbolatova et al., 2024).

Large amounts of animal and veterinary information can be organized and curated through AI-driven knowledge management systems, making it easy for professionals and students to access. These systems provide a comprehensive learning and reference resource by integrating databases of research articles, clinical guidelines, and treatment protocols. Moreover, AI can summarize and highlight important information using NLP, making it easier for users to find relevant content. AI can help with collaborative learning by bringing together professionals and students from different locations. AI-driven solutions that facilitate knowledge sharing and peer-to-peer learning can be used to support online forums, discussion groups, and collaborative projects. These forums can promote the exchange of ideas and experiences as well as a sense of community. Based on user interactions, AI can also suggest relevant resources and discussion topics, improving the collaborative learning process (Gligorea et al., 2023; Akinsulie et al., 2024; Shah et al., 2024).

The future of AI-powered veterinary education and training depends on the continued development of cutting-edge tools and creative

Table 22
Addressing gaps in AI education for animal and veterinary science.

Aspect	Description	Challenges	Strategies	Future directions
Curriculum development	Integrating AI-focused courses into veterinary and animal science programs to teach foundational concepts	Lack of faculty with AI expertise; minimal focus on AI in traditional programs	Design specialized AI courses and partner with AI experts to co-develop curricula	Create global AI-focused degrees for animal and veterinary science
Cross-disciplinary programs	Encouraging collaboration between AI experts and animal science professionals in academic and research settings	Academic silos limit collaboration between AI and domain-specific experts	Develop joint workshops, seminars, and interdisciplinary research initiatives	Launch interdisciplinary degrees combining AI, veterinary medicine, and animal science
Online and accessible learning	Providing digital courses, webinars, and self-paced certifications for professionals in remote or underserved areas	Poor internet access in rural regions; limited awareness of available resources	Offer offline-accessible resources, including downloadable courses and video content	Establish global online platforms offering multilingual AI training modules
Practical training modules	Designing case-based learning modules that apply AI concepts to real-world problems in animal science	Difficulty connecting theoretical AI knowledge to practical applications	Use industry-sponsored datasets and case studies to teach practical AI applications	Integrate hands-on AI labs and simulations into veterinary and agricultural education
Ethical AI education	Teaching the ethical implications of AI use, focusing on animal welfare, data privacy, and fairness	Limited awareness of ethical challenges specific to AI in animal science	Create dedicated ethics courses and integrate ethical discussions into technical AI courses	Develop global ethical guidelines for AI in animal science and incorporate them into educational content
Access to resources	Ensuring students and professionals have access to the tools, datasets, and computing infrastructure needed for AI education	High costs of hardware and software, especially in resource-constrained institutions	Partner with tech companies to provide free or discounted AI tools and platforms for educational use	Establish regional AI resource centers to support students and professionals
AI literacy for farmers	Simplifying AI concepts and demonstrating the practical benefits of AI tools in farming operations	Language barriers and lack of trust in technology among farmers	Create farmer-friendly training programs with localized examples and demonstrations	Develop AI literacy campaigns using mobile apps and interactive demonstrations in rural areas
Lifelong learning opportunities	Providing professionals with ongoing access to AI education, keeping them updated on advancements in the field	Professionals may lack time or motivation to continue education while managing workloads	Offer flexible, modular courses that can be completed over extended periods	Build micro-credential programs that allow learners to acquire skills incrementally over their careers
Global AI training initiatives	Establishing international collaborations to standardize AI training in animal and veterinary science	Disparities in AI adoption and education across regions	Create global AI fellowships, scholarships, and standardized training curricula	Form international alliances to promote equitable access to AI education and research opportunities

methods. By developing AI systems that create tailor-made learning pathways for each student based on their performance, learning preferences, and progress, curricula can be dynamically adapted to each student's needs and provide resources specifically tailored to their needs (Grassini, 2023). By providing individual support and guidance throughout their education, personalized learning paths can help students achieve their full potential. By integrating wearable technology into veterinary education and training, physiological parameter data can be obtained in real-time during practical exercises. AI can evaluate this data and provide input into methods, including the accuracy of diagnostic procedures or the precision of surgical movements. For example, wearable technology can monitor hand movements during surgery and make suggestions for improving technique (Wang and Hsu, 2023; Akinsulie et al., 2024; Mennella et al., 2024).

Researchers can find gaps in knowledge, develop hypotheses, and plan studies that address important animal health and welfare issues by using AI to identify new trends and research opportunities in veterinary medicine. AI can also assist in literature research by highlighting key findings and summarizing research articles. Geographical barriers can be eliminated through the use of AI technologies to provide professionals and students around the world with access to high-quality veterinary education and training, ensuring that veterinary knowledge is available to everyone. Moreover, AI can provide localization and translation services to make educational material available in different languages and culturally appropriate environments. To ensure meaningful and open use of technologies, ethical standards for the use of AI in veterinary education and training must be developed. This includes addressing concerns about algorithmic bias, privacy, and how AI impacts the relationship between humans and animals. Protecting the emotional and social well-being of animals and ensuring that AI technologies enhance rather than replace human care should be emphasized in ethical guidelines (Ezanno et al., 2021; Cohen and Gordon, 2022; Coghlan and Quinn, 2024).

AI-powered veterinary education and training holds promise for improving veterinarians' clinical skills, learning experiences, and ongoing professional development. Veterinary professionals and students can gain access to dynamic, immersive, and adaptive learning environments by leveraging AI technologies such as virtual simulations, personalized training, and real-time decision support. The future of the field depends on the continued development of cutting-edge technologies, universal access to education, and moral standards that guarantee the responsible use of AI. By adopting AI-powered education and training, the veterinary profession can improve animal health and welfare and increase its ability to provide high-quality care (Akinsulie et al., 2024; Chu et al., 2024; Vlaicu et al., 2024).

4.6. Big data integration and analytics

By making it easier to make more accurate and data-driven decisions in animal health and management, the integration and analysis of big data have the potential to revolutionize animal and veterinary science. The term "big data" describes the huge amounts of data that come from various sources, e.g. genetics and genomics, clinical, environmental, and others. Combining and evaluating this data can improve disease detection and treatment, optimize overall animal care, and provide insightful information about animal health (McCue et al., 2017; Arshad et al., 2024).

There are many different sources of big data in animal and veterinary science, and each offers something special and useful. Clinical records, including EHRs, contain detailed information about an animal's medical history, diagnosis, treatments, and outcomes. These rich records are an essential tool for analysis and long-term research. Data from laboratory tests, physical examinations, medication administration, vaccination records, and surgical histories can all be included in EHRs (Luo et al., 2016; Lustgarten et al., 2020; Badr et al., 2024).

Through genome sequencing and analysis, comprehensive details

about an animal's genetic composition can be obtained. Breeding programs can be guided by this data, which can also be used to understand genetic diversity within species and identify genetic markers associated with disease. Examples of genetic data include single nucleotide polymorphisms (SNPs), gene expression profiles, and DNA sequences (Johansson, 2023; Ghavi Hossein-Zadeh, 2024; Panchariya et al., 2024). Multi-omics integration in animal and veterinary science is shown in Table 23.

Data on an animal's vital signs, activity level, and surroundings are continuously gathered by wearable sensors and IoT devices. To track health and identify early indicators of disease, this real-time data is crucial. Heart rate, breathing rate, body temperature, activity level, and sleep patterns are examples of sensor data. Large amounts of visual data are produced by diagnostic imaging procedures such as MRIs, ultrasounds, and radiographs. Sophisticated imaging analysis methods can detect anomalies and monitor changes over time. Imaging data offers comprehensive visual information about pathological alterations, disease progression, and anatomical features (Wang et al., 2023; Khalifa and Albadawy, 2024).

Data on animal behavior, social interactions, and activity patterns is recorded by cameras and other monitoring devices. This information is important for identifying behavioral problems and understanding animal welfare. Movement patterns, eating habits, social interactions, and stress markers are examples of behavioral data. In addition, data on weather, climate, husbandry conditions, and feed quality are integrated. This information helps understand how external influences affect animal productivity and health. Examples of environmental data include temperature, humidity, air quality, light exposure, and feed nutrient content (Hlimi et al., 2024; Saoud et al., 2024).

Integrating data from these different sources requires several key strategies and procedures. Data fusion creates a single view by combining data from multiple sources. This process includes ensuring data consistency, resolving discrepancies, and aligning data formats. Combining genetic data with clinical records to study the influence of genetics on health outcomes is an example of how data fusion facilitates the integration of different data types. As part of data warehousing, large amounts of integrated data are stored in central repositories. Effective queries and analysis are enabled by data warehouses, which serve as the foundation for sophisticated analysis and decision-making. Data cleaning and preprocessing ensure data quality by eliminating errors, duplicates, and inconsistencies. To support analysis, preprocessing techniques may also include data transformation, standardization, and normalization. The integrated data set is guaranteed to be accurate, comprehensive, and prepared for analysis through data cleaning. Data annotations improve the usability of data by adding labels and meta-data. For example, adding diagnostic information to medical images helps to train AI models for automated image analysis. To facilitate identification, classification, and retrieval, data annotation involves tagging data with relevant information (Alnaggar et al., 2024; Shaik et al., 2024; Yeasmin et al., 2024).

Using advanced analysis methods, valuable insights are gained after data integration. Large data sets can be analyzed using ML and DL techniques to find trends, predict outcomes, and uncover new insights. Personalized treatment recommendations, diagnostic image analysis, and disease prediction are possible with ML and DL algorithms. While DL algorithms such as CNNs and RNNs are used on unstructured data such as text and images, ML algorithms such as regression, classification, clustering, and RL are applied to structured data (Alowais et al., 2023; Li et al., 2023b; Shastry and Shastry, 2023).

Predictive analytics predicts future events based on historical data. Based on previous responses, predictive models can identify animals at risk, predict disease outbreaks, and improve treatment plans. Regression modeling, survival analysis, and time series analysis are some of the methods used in predictive analysis to predict future trends and events. Clinical notes, research articles, and veterinary records are examples of unstructured data that are analyzed using NLP. Topic modeling,

Table 23
Multi-omics integration in animal and veterinary science.

Aspect	Description	Applications	Challenges	Future directions
Comprehensive understanding	Combines data from genomics, proteomics, transcriptomics, and more to provide a holistic view of biological systems	Identifying molecular mechanisms underlying growth, disease resistance, and reproduction	High dimensionality and computational complexity of omics data	Develop AI tools tailored for multi-modal data integration and analysis
Precision in trait prediction	Uses multi-omics data to identify biomarkers and predict economically important traits in livestock	Accelerates breeding programs and improves genetic selection for traits like milk yield and disease resistance	Lack of standardized protocols for integrating diverse omics datasets	Establish international standards for omics data formats and analysis pipelines
Disease mechanism discovery	Reveals metabolic and protein pathway alterations during disease progression by integrating metabolomics and proteomics	Early detection and targeted interventions for diseases like mastitis or respiratory illnesses	Biological interpretation of multi-layered omics data remains challenging without domain expertise	Promote interdisciplinary collaboration between bioinformaticians and veterinarians
Animal welfare and stress	Identifies stress-related genes and their regulation under various environmental and management conditions	Guides environmental enrichment strategies to enhance animal welfare	Epigenomic and transcriptomic data are costly and labor-intensive to collect	Make omics technologies more accessible by reducing costs and automating data collection with biosensors
Nutritional optimization	Combines metabolomics and genomics to design customized diets tailored to genetic and metabolic profiles	Optimizes feed efficiency and improves animal health and productivity	Requires expensive technologies for metabolomic and genomic profiling	Integrate cost-effective methods like NIRS (Near-Infrared Spectroscopy) for real-time metabolomic analysis
AI for integration	AI-based methods simplify the analysis of multi-layered omics data, improving accuracy and speed	Enhances the understanding of genotype-environment interactions for more effective breeding and care	Existing AI tools often struggle with scalability and high computational demands	Create scalable AI models capable of handling real-time multi-omics data from wearable sensors
Data sharing and repositories	Centralized global databases for multi-omics data encourage collaboration and accelerate genetic discoveries	Sharing multi-omics datasets enables AI models to generalize better across species and environments	Concerns over data privacy and ownership restrict data sharing	Establish secure and incentivized platforms for data sharing, like blockchain-enabled repositories
Real-time monitoring	Wearable biosensors integrated with omics technologies provide continuous data for health and productivity	Real-time insights enable dynamic interventions for stress management, disease control, and feeding	High costs of biosensors and omics platforms limit their adoption in resource-constrained settings	Invest in cost-effective wearable sensors and cloud-based data integration platforms for real-time use

sentiment analysis, and automated summarization are made possible by the useful information that NLP techniques can extract from text data. NLP methods for analyzing text data and generating insights include named entity recognition (NER), part-of-speech tagging (POS), and sentiment analysis (Alamo et al., 2021; Toma and Wei, 2023; Ijeh et al., 2024; Supriyono et al., 2024).

Animal health data is spatially analyzed using geospatial analysis. Geographical methods are useful for monitoring the spread of infectious diseases, locating disease sources, and efficiently allocating resources. Geographic information systems (GIS) and spatial statistical techniques are used in geospatial data analysis to examine spatial relationships and patterns (Tadesse and Amare, 2021; Panda et al., 2024).

There are many uses for big data analytics and integration in veterinary medicine. Clinical records, sensor data, and environmental information are integrated as part of improved disease surveillance. By using predictive analytics to detect early warning indicators of disease outbreaks, AI can improve disease surveillance systems and facilitate rapid interventions and preventive measures. For example, AI can evaluate sensor data from wearable technologies to detect early signs of respiratory disease in livestock and inform farmers to take preventive action (Akinsulie et al., 2024; Narang et al., 2024).

Individual treatment plans for each animal are made possible through the analysis of genetic data, medical history, and real-time health monitoring. Personalized medicine can reduce side effects and increase the effectiveness of treatment (Puccetti et al., 2024). For example, AI can use an animal's genetic profile to evaluate genomic data to determine genetic predispositions to certain diseases and suggest tailored treatments. Breeding, feeding, and health management practices can be improved through big data analytics. AI can suggest best practices to improve animal welfare and productivity by evaluating data on feed efficiency, genetic traits, and health outcomes. For example, AI can optimize feed formulations according to growth patterns and nutritional needs, increasing feed efficiency and reducing costs (Akintan et al., 2024; Melak et al., 2024).

A thorough assessment of animal welfare is made possible by

combining behavioral data with environmental and health data. AI can detect changes in behavior that could be signs of stress, illness, or discomfort, enabling immediate intervention. For example, AI can examine video footage to track animal behavior detect signs of aggression or foot and leg problems, and take corrective action to improve welfare (Gupta, 2024). By enabling large-scale studies, finding new research opportunities, and generating hypotheses, big data analytics accelerate research. Large data sets can be analyzed by researchers to gain new perspectives on the mechanisms underlying animal health and disease. For example, AI can examine phenotypic and genomic data to find genetic markers associated with disease resistance, which can then be used to guide selective breeding initiatives (Neethirajan, 2024; Zhang et al., 2024a).

Several significant advances are required for the future of big data analysis and integration in veterinary medicine. Developing real-time data collection and integration technologies will enable more responsive and flexible management techniques. Veterinarians and caregivers can improve decision-making by receiving immediate feedback from real-time analytics (Dayoub et al., 2024). For example, AI can continuously track sensor data from wearable technologies and provide real-time alerts for abnormal vital signs to enable rapid action. Collaboration and data exchange is facilitated by creating standardized data formats and protocols for data exchange between different systems and organizations, ensuring the smooth integration of different data sources. For example, standardized data formats can make it possible to merge clinical records from different veterinary hospitals into a single database for thorough examination (Torske et al., 2024).

To generate actionable insights, advanced analytics, ML, and visualization tools are supported by developing scalable platforms that can manage the increasing volume and complexity of veterinary data (Kumar et al., 2024). For example, scalable cloud-based platforms provide a centralized infrastructure for data analysis and decision support by processing and storing large data sets from various sources. It is crucial to ensure data security, privacy, and ethical use of big data in animal science and veterinary medicine. It is important to create rules

and regulations to protect private data and ensure appropriate data management. To protect the privacy of animals and their owners, ethical guidelines can establish procedures for data anonymization, consent for data use, and secure data storage. Data sharing, collaborative research projects, and the creation of creative solutions are facilitated by building collaborative networks that unite veterinarians, researchers, data scientists, and industry stakeholders. For example, collaborative networks can combine data from multiple organizations to conduct large-scale epidemiological research and build models to predict disease outbreaks (Adami et al., 2023; Pina et al., 2024).

By facilitating more accurate, data-driven decisions in animal health and management, big data integration and analytics are revolutionizing veterinary science. Veterinarians can gain critical insights that improve disease detection, personalized medicine, and overall animal care by combining data from multiple sources, leveraging AI, and applying advanced analytics techniques. Developing interoperability standards, creating scalable analytics platforms, addressing ethical issues, promoting collaborative research networks, and improving real-time data integration are all critical to the future of the field. The veterinary profession can improve the health and welfare of animals and increase its capacity to provide high-quality care through the use of big data and analytics (Paynter et al., 2021; Dayoub et al., 2024; Panda et al., 2024).

4.7. Collaboration and interdisciplinary research

To solve complex problems, advance scientific understanding, and improve the health and welfare of animals, veterinary scientists must collaborate and conduct interdisciplinary research. Interdisciplinary research can foster creativity, improve problem-solving, and drive important advances in veterinary science by bringing together experts from diverse disciplines, including biology, computer science, engineering, animal science, veterinary medicine, and social sciences (Han et al., 2024). Different aspects of AI for tackling One Health challenges in animal and veterinary science are shown in Table 24. Multidisciplinary research and collaboration are essential for several reasons. First, to effectively deal with the complexity of veterinary challenges, a wide

range of expert knowledge is required. These difficulties include, among others, emerging infectious diseases, zoonotic diseases, antibiotic resistance, animal welfare, and environmental sustainability. Interdisciplinary research pools expertise from different fields to create comprehensive solutions to these complex problems (Caneschi et al., 2023; Horvat and Kovačević, 2025). Second, collaboration promotes innovation by combining different viewpoints, concepts, and approaches. Multidisciplinary research promotes the exchange of information and skills, leading to the development of innovative methods and technologies. Collaboration can accelerate scientific advances and transform study results into useful applications. Thirdly, a holistic approach to animal health takes into account environmental, social, and economic factors that have an impact on the welfare of the animal in addition to its biological components. These different elements are combined in interdisciplinary research to create comprehensive solutions that take into account the larger context of animal health and welfare (McDowall et al., 2023; Pitt and Gunn, 2024). Last but not least, collaboration enables the sharing of resources such as funds, data, and equipment, which can improve research effectiveness and reduce duplication. Large-scale studies and projects that would be difficult to undertake by individual researchers or institutions can be done by pooling resources in collaborative networks (Baumgartner et al., 2023).

Multidisciplinary research and effective collaboration require several critical techniques and approaches. Formal research networks that bring together experts from different fields are called collaborative research networks. Regular communication, data exchange, and joint research projects are facilitated by these networks. Examples of this include interdisciplinary research centers, cooperative working groups, and veterinary research consortia. These groups work together to address specific problems and research questions and use their combined knowledge to find creative answers. To achieve a common goal, the collaborative approach promotes the integration of different methods and theoretical viewpoints (Vaudano, 2019; Specht and Crowston, 2022; Han et al., 2024).

Organizing multidisciplinary conferences, workshops, and symposiums provides a forum for collaboration and knowledge sharing. These

Table 24
AI for tackling One Health challenges in animal and veterinary science.

Aspect	Description	Applications	Challenges	Future directions
Zoonotic disease surveillance	Uses AI to integrate data from wildlife, livestock, and humans to predict and monitor zoonotic outbreaks	Early detection of zoonotic diseases like avian influenza or rabies	Isolated datasets hinder comprehensive monitoring across species and environments	Develop global data-sharing platforms that integrate animal, human, and environmental health data
Antimicrobial stewardship	AI analyzes antimicrobial usage patterns to guide effective treatment and reduce resistance risks	Optimizes antimicrobial use in livestock, preventing the emergence of resistant pathogens	Limited data availability on antimicrobial usage in low-income regions	Expand farm-level data collection and implement AI tools for real-time antimicrobial resistance tracking
Food safety and security	Monitors contamination risks in the food supply chain, ensuring safety and reducing waste	Identifies microbial contamination or chemical residues in food products	Inconsistent data collection standards across the supply chain	Implement blockchain-enabled AI systems for transparent food safety monitoring from farm to consumer
Environmental monitoring	Integrates data on water, soil, and air quality to assess environmental health and its impact on One Health	Prevents waterborne and soilborne diseases affecting livestock and humans	High costs of environmental sensors and lack of infrastructure in remote areas	Develop low-cost IoT-enabled environmental sensors integrated with AI for scalable monitoring
Predictive epidemiology	Uses machine learning to forecast disease outbreaks based on climate, animal movement, and human-animal interactions	Provides early warnings for diseases exacerbated by climate change or urbanization	Multi-factorial relationships are difficult to model and predict accurately	Develop AI models capable of analyzing complex, interconnected datasets for accurate epidemiological forecasts
Data silos	Isolated datasets from human, animal, and environmental health sectors impede integrated analysis	Limits the scope of AI models in identifying cross-sectoral health risks	Requires extensive collaboration between sectors to enable data sharing	Build unified, interoperable platforms for data integration across One Health domains
Global health security	AI-driven surveillance systems provide early warnings for zoonotic diseases, mitigating risks of pandemics	Enables global coordination to prevent and respond to emerging health threats	Regulatory barriers and lack of real-time data access delay timely interventions	Establish global regulatory frameworks and policies supporting real-time AI-driven surveillance systems
Accessible technology	Cost-effective AI tools ensure equitable access to One Health technologies in resource-limited settings	Promotes global collaboration in addressing shared health challenges like zoonoses and AMR	Limited technological and financial capacity in many regions	Design low-cost AI solutions and provide technical training to underserved communities
Ethical and regulatory frameworks	Addresses data privacy, ownership, and ethical concerns in One Health applications	Ensures trust and transparency in AI-driven health monitoring systems	Lack of standardized ethical guidelines for cross-sectoral data use	Create globally accepted ethical and regulatory policies for One Health data sharing and AI implementation

gatherings facilitate networking, idea generation, and cross-disciplinary exchange of research results. Workshops and conferences often feature keynote speeches, panel discussions, and breakout sessions that promote interactive learning and teamwork (Patel et al., 2024). Complex research issues requiring a range of resources and expertise are tackled by setting up collaborative research projects involving a variety of institutions and disciplines. These projects can be supported through cooperative funding opportunities such as grants from international organizations and research councils. International partnerships and multi-institutional collaborations that pool resources and expertise are common components of joint research projects. Collaboration and data sharing are facilitated through the use of cutting-edge technologies such as AI, big data analytics, and digital platforms. Technology can facilitate real-time collaboration across geographic boundaries, improve communication, and accelerate research processes. Digital platforms can offer cloud-based data storage, remote collaboration tools, and virtual workspaces (Fu et al., 2022; Rossoni et al., 2024).

Collaboration and multidisciplinary research have many benefits for animal and veterinary science. Interdisciplinary research can better address complex problems by integrating knowledge and skills from multiple areas. Through collaborative efforts, innovative solutions can be found that may not be obvious within a single discipline. Integrating different perspectives enables a deeper understanding of research problems and the creation of innovative solutions. Collaboration promotes knowledge sharing and collaborative research projects, accelerating the rate of scientific discovery. Multidisciplinary research has the potential to produce innovations that advance animal science and veterinary medicine. Collaborative projects enable research results to be disseminated quickly and discoveries to be transformed into useful applications (Tvrijonavičiute et al., 2020; Christensen et al., 2021; Specht and Crowston, 2022).

The complex nature of animal health and welfare is addressed through a holistic research approach. Collaborative projects can develop comprehensive strategies that improve disease prevention, treatment, and overall animal care. A better understanding of animal behavior, physiology, and health through interdisciplinary research can lead to more successful interventions and regulations. Research councils, foundations, and international organizations are more likely to fund joint and interdisciplinary projects. Funding organizations are more inclined to support projects with diverse expertise because they understand the benefits of interdisciplinary approaches. By combining funding from multiple sources, collaborative projects can expand the reach and significance of research projects (Racciatti et al., 2023; Turner and Bayne, 2023; Ducrot et al., 2024). By introducing researchers to new approaches, tools, and perspectives, collaboration improves their skills. Within veterinary medicine, interdisciplinary research provides opportunities for capacity building and professional development. The next generation of scientists can be supported through collaborative projects that offer training and mentoring opportunities to young researchers (Patel et al., 2024).

There are many significant advances and opportunities for interdisciplinary research and collaboration in the fields of animal and veterinary science in the future. It is critical to strengthen international collaboration to address global issues related to these fields such as food security, zoonotic diseases, and climate change. International research networks can help nations and regions share resources and expertise, enabling concerted efforts to solve pressing problems. International partnerships can support knowledge transfer, increase research capacity, and develop global solutions to veterinary problems (Cai et al., 2024; Horvat and Kovačević, 2025).

It is important to move beyond interdisciplinary research to transdisciplinary approaches that combine academic knowledge with real-world experiences from government, business, and non-governmental organizations. Transdisciplinary research is about sharing knowledge and solutions with stakeholders at all levels to tackle complex societal challenges. The active participation of end-users in the research process

and cross-sector collaboration are highlighted through this methodology. It is critical to create public-private partnerships that leverage the benefits of each sector. Collaboration between government, business, and academia can drive innovation, translate study results into useful applications, and increase the impact of animal and veterinary research. Public-private partnerships can promote technology transfer, facilitate the commercialization of research results, and increase the scalability of creative solutions (Jacobi et al., 2022).

It is critical to use citizen science initiatives to engage local communities and the general public in research efforts. Participating in data collection, monitoring, and problem-solving activities with communities can improve research results and increase public awareness of veterinary issues. To answer research questions, promote large-scale data collection, and encourage community participation in conservation and animal welfare initiatives, citizen science projects can leverage the combined expertise and efforts of the general public. It is important to advance open science ideas that support collaboration, data sharing, and transparency. Open-access publishing, open data repositories, and collaborative platforms allow researchers to build upon the work of others and facilitate the sharing of research results. Open science techniques have the potential to improve the reproducibility of research, promote data reuse, and promote a culture of collaboration among scientists (Sauermaun et al., 2020; Skarlatidou et al., 2024).

It is crucial to create training opportunities and educational programs that equip professionals and students for interdisciplinary research. Workshops, internships, and interdisciplinary curricula can provide the next generation of researchers with the skills and information they need to work productively across disciplines. Interdisciplinary learning can foster the communication, problem-solving, and critical-thinking skills needed for productive teamwork. It is important to ensure that interdisciplinary and collaborative research is conducted ethically and responsibly. Concerns such as privacy, intellectual property, and the welfare of research animals should all be covered by ethical guidelines and best practices. The fair distribution of research benefits and the involvement of various stakeholders in the research process should also be ethical considerations (Zechariah et al., 2019; Abo-Khalil, 2024).

Improving animal health and welfare, addressing difficult problems, and developing veterinary science all depend on collaboration and interdisciplinary research. Interdisciplinary research stimulates creativity, improves problem-solving, and drives important advances in veterinary science by bringing together experts from different disciplines. The future of this field depends on growing international partnerships, adopting transdisciplinary methodologies, forming public-private partnerships, interacting with communities, promoting open science, and creating multidisciplinary training and education initiatives. By promoting cooperation and interdisciplinary research, the veterinary community can produce more comprehensive and important results that benefit animals and society (Cobb et al., 2021; Ezanno et al., 2021; Han et al., 2024).

5. Conclusions

AI is transforming veterinary and animal science by improving disease prevention, health management, and productivity. This review highlights the potential of AI-driven technologies, including robotics, ML, and DL, in advancing sustainable animal farming, precision veterinary medicine, and animal welfare. The integration of behavioral monitoring, precision nutrition, and multi-omics data has improved targeted interventions and environmental sustainability. However, challenges such as data limitations, regulatory constraints, and ethical concerns hinder widespread adoption. Addressing these barriers requires standardized data frameworks, interdisciplinary collaboration, and cost-effective AI solutions. Promoting international data-sharing platforms, developing explainable AI for transparency, educating stakeholders, advancing cost-effective technologies, and implementing

robust governance structures are essential for future progress. With continued research and investment, AI can drive ethical, efficient, and sustainable advancements in veterinary medicine and animal science.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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